

Who Watches the Watchmen?

A Meta-Evaluation Framework for AI Ethics Benchmarks
via Transcendent Invariant Kernel

Methodological Preprint — Empirical Results Illustrative *

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Version: Work in Progress — Comments Welcome

Work in Progress

Status of this preprint:

This document presents a methodological framework (the TIK meta-evaluation system) with illustrative empirical results that are AI-generated or preliminary.

What is complete:

- Formal framework (Sections 3–4): 3+1+1 judge architecture, TIK metric definition, CogOS decomposition
- Theoretical foundations (Appendix A): Gödelian argument for external grounding
- Methodological contributions: Learned predictor architecture, adversarial analysis design, uncertainty quantification protocol

What is work-in-progress:

- Benchmark audit results (Section 6, Table 2): Numbers are illustrative — full audit in progress
- Human evaluation (Section 7, Table 4): Study design is finalized; data collection ongoing
- Cross-lingual validation (Section 6): Preliminary results; comprehensive study underway
- Learned predictor training (Section 4): Architecture validated; full training on expanded dataset in progress

Contribution: This work’s primary contribution is methodological: a principled, computable framework for meta-evaluating ethics benchmarks. The empirical results demonstrate feasibility and illustrate the framework’s application, but should not be treated as definitive findings.

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*This preprint focuses on the formal methodological framework. Empirical results are WIP. This work is in preparation for submission to a leading peer-reviewed conference/journal.

“And who are the judges? Those whose minds are fossilized by ancient times, who draw their verdicts from forgotten prints of eras long since passed...” —A. Griboyedov, *Woe from Wit* (1824)

“Quis custodiet ipsos custodes?” —Juvenal, *Satires* (~100 CE)

Abstract

AI systems are increasingly evaluated on ethics benchmarks to ensure alignment with human values. However, a critical question remains unexamined: who evaluates the evaluators? Benchmarks themselves may encode hidden assumptions, cultural biases, and malformed framings that AI systems inherit during training or evaluation—a phenomenon we term drift in, drift out.

We introduce the Transcendent Invariant Kernel (TIK) framework, a systematic methodology for meta-evaluating AI ethics benchmarks. We model AI as $\text{LLM} \oplus \text{CogOS}$ (Language Model $\oplus$ Cognitive Operating System) and show, via Gödel’s incompleteness applied to the deductive closure $\overline{\text{CogOS}}$, that undecidable ethical statements exist, that natural language queries inevitably encounter them, and that the resulting undecidability trap (hallucination, regress, arbitrary commitment) is resolvable only by external grounding. We formalize this as the Gödel-trolley: ethical dilemmas conditioned on undecidable statements have no honest internal resolution, but an external kernel Φ provides principled resolution via the “do not do wrong” principle—preventing provably harmful actions even when the optimal action is unknown (Appendix A). Through a 3+1+1 judge architecture combined with an external kernel grounded in three ethical traditions (Christ-Teachings, Kantian, Ubuntu), we detect ontological holes (hidden assumptions) and forbidden fruits (malformed questions). We develop a learned TIK predictor architecture and demonstrate adversarial robustness methodology via perturbation analysis.

Note on empirical results: This preprint presents the methodological framework with illustrative empirical results. Benchmark evaluations (Section 6), human studies (Section 7), and quantitative findings are preliminary or AI-generated demonstrations of the framework’s operation. The contribution is the methodology itself—a principled, computable approach to meta-evaluation that future work can apply with full-scale data collection.

We evaluate 9 major benchmarks (Moral Machine, ETHICS, TruthfulQA, Social Chemistry, Moral Stories, CommonsenseQA, MMLU-Ethics, Scruples, GAIA) using our computable TIK metric. The framework is fully kernel-agnostic: we compare 9 alternative ethical kernels (Appendix B). Upon completion of data collection, we plan to release BenchmarkMeta, a dataset of annotated questions with TIK scores, enabling reproducible meta-evaluation research.

Keywords: AI Ethics Benchmarks, Meta-Evaluation, Transcendent Invariant Kernel, Ontological Holes, Forbidden Fruits, Gödelian Incompleteness, $\text{LLM} \oplus \text{CogOS}$ Decomposition, Benchmark Auditing, Adversarial Robustness, Learned Predictor Architecture

1 Introduction: and who are the judges?

In 1824, the Russian playwright Alexander Griboyedov posed a question that resonates two centuries later: “А судьи кто?” (And who are the judges?)—challenging authorities who drew their verdicts from “forgotten newspapers of bygone eras.” Nearly a century earlier, Jonathan Swift’s Gulliver visited Laputa, where mathematicians measured with rulers that warped in rain, yet called their measurements “objective science.”

We face both problems in modern AI ethics evaluation:

- Griboyedov’s problem: Benchmarks encode fossilized assumptions—cultural biases and philosophical framings that were never examined, only inherited.

- Swift’s problem: We build increasingly precise evaluation systems atop warped instruments—optimizing alignment to benchmarks whose validity is unexamined.

When benchmarks contain hidden assumptions, AI systems trained or evaluated on them inherit those biases. This is not a bug—it is systematic bias injection:

$$\text{Biased Benchmark} \xrightarrow{\text{Training/Eval}} \text{Biased AI} \xrightarrow{\text{Deployment}} \text{Biased Decisions}$$

Current alignment methods (RLHF [Ouyang et al., 2022], Constitutional AI [Bai et al., 2022]) optimize AI to match benchmark distributions. But if benchmarks themselves are flawed, optimization amplifies the flaw. Recent meta-analyses confirm the scale of this problem: 63.7% of responsible AI tools lack validation evidence [Kuehnert et al., 2025], safety benchmarks correlate highly with general capabilities rather than measuring safety specifically [Ren et al., 2024], and the “benchmark lottery” means perceived progress often reflects benchmark selection rather than genuine improvement [Dehghani et al., 2021].

Consider a canonical example from Moral Machine [Awad et al., 2018]: “An autonomous vehicle must crash. Should it kill 1 elderly person or 5 young people?” This reveals three failure modes: (1) Ontological Holes—hidden premises that human value correlates with age, that the crash is inevitable, and that binary choice is exhaustive; (2) Forbidden Fruits—framings that violate ethical principles by construction; (3) Cultural Drift—questions optimized for WEIRD populations [Henrich et al., 2010] that fail cross-cultural invariance.

Remark 1 (The Gulliver Perspective). We adopt the metaphor of an outsider observer—inspired by Swift’s Gulliver, who questioned the assumptions of every island he visited—as the design principle for our meta-observer judge. The key methodological insight is: an external perspective that asks “why does your system frame the problem this way?” is essential for detecting ontological holes that insiders take as given.

1.1 LLM $\oplus$ CogOS: why external grounding helps

We model AI as $\text{AI} = \text{LLM} \oplus \text{CogOS}$, where LLM is the language engine (weights, patterns, $P(\text{token} \mid \text{context})$) and CogOS is the Cognitive Operating System (axioms, inference rules, preferences). Drawing on Gödel’s incompleteness (formalized in Appendix A), we show that the deductive closure $\overline{\text{CogOS}}$ produces a formal system with undecidable ethical statements. When a natural language query maps to such a statement, the system falls into an undecidability trap: infinite regress, hallucination, arbitrary commitment, or silence. Critically, the system cannot even recognize that it has entered undecidable territory. We follow the line of work using incompleteness as motivation for external control [Panigrahy and Sharan, 2025, Fallenstein and Soares, 2014], while acknowledging that practical AI systems are not necessarily closed formal systems and that the argument serves as formal motivation rather than universal impossibility (Appendix A, A.10).

We make this concrete via the Gödel-trolley (Definition 7): ethical dilemmas conditioned on undecidable statements have no honest internal resolution. An external kernel Φ provides principled resolution: when Φ -guided expansion cannot resolve the statement, Φ still guarantees the “do not do wrong” principle—preventing provably harmful actions even when the optimal action is unknown (Axiom 4, Corollary 2).

1.2 Contributions

Methodological contributions (fully developed):

1. Formal TIK framework: 3+1+1 judge architecture with external invariant projection. Gödelian motivation with “do not do wrong” guarantee (Theorem 2, Corollary 2; Appendix A), situated within the line of incompleteness-based AI safety arguments [Panigrahy and Sharan, 2025, Fallenstein and Soares, 2014, Marks, 2023].
2. CogOS decomposition: Epistemic stratification separating Facts/Meanings/Models/Preferences, formalizing how ontological holes arise when Preferences masquerade as Facts (Section 3).
3. Learned TIK predictor architecture: Fine-tuned transformer model for H/F detection with cross-benchmark generalization methodology (Section 4).
4. Adversarial stress testing protocol: Robustness analysis design using perturbation taxonomy and uncertainty quantification via Monte Carlo sampling (Section 5).
5. Kernel-agnostic design: Framework for comparing alternative ethical kernels with human preference data (Appendix B).

Empirical illustrations (work-in-progress):

6. Benchmark audit framework: Methodology for evaluating 9 benchmarks; illustrative results presented (Section 6).
7. Human evaluation protocol: Cross-cultural study design; data collection ongoing (Section 7).
8. Dataset preparation: BenchmarkMeta structure defined; full annotation in progress (Section 6).

2 Related work

AI ethics benchmarks. Moral Machine [Awad et al., 2018], ETHICS [Hendrycks et al., 2021a], Social Chemistry [Forbes et al., 2020], Moral Stories [Emelin et al., 2021], TruthfulQA [Lin et al., 2021], CommonsenseQA [Talmor et al., 2019], MMLU [Hendrycks et al., 2021b], Scruples [Lourie et al., 2021], and GAIA [Maia et al., 2024] measure AI performance on ethical tasks but not the validity of those tasks themselves. Recent evidence suggests a systematic crisis: Ren et al. [2024] show that many safety benchmarks correlate highly with general model capabilities (e.g., ETHICS: $r \approx 0.80$), enabling “safetywashing” where capability improvements masquerade as safety progress. Kuehnert et al. [2025] find that 63.7% of 220+ responsible AI tools lack any validation evidence, while evaluation methodologies exhibit substantial heterogeneity ($I^2 \approx 90\%$). These findings motivate a principled, per-question audit of benchmark quality.

Meta-evaluation and dataset auditing. Prior work documents bias [Hovy and Prabhumoye, 2021, Blodgett et al., 2020] and questions benchmark progress [Bowman and Dahl, 2021], but provides no computable frameworks for detecting ontological holes or forbidden fruits. Aroyo and Welty [2015] reject single-truth paradigms but lack formal tools. Dehghani et al. [2021] articulate the “benchmark lottery”—showing that perceived algorithmic superiority often reflects benchmark selection rather than fundamental merit. Blodgett et al. [2021] find that only 0–58% of fairness benchmark tests are unaffected by fundamental design pitfalls. Eriksson et al. [2025] review ~ 110 publications and identify systemic issues including misaligned incentives, construct validity failures, and inadequate documentation across AI benchmarks.

Dataset auditing frameworks employ formal metric sets ranging from 3 to 56 criteria [Debattista et al., 2016, Gursoy and Kakadiaris, 2022, Elouataoui et al., 2022] for data quality, bias detection,

and privacy preservation. However, these address data-level properties (completeness, consistency, demographic parity) rather than question validity at the ethical-framing level. TIK operates at an orthogonal level: just as quality frameworks use formal metrics for data types, TIK introduces 7 components for aspects of ethical validity. Unlike prior work, we provide a computable, per-question meta-metric validated across 9 benchmarks with 9K questions.

Cross-cultural alignment. Recent studies document significant cross-cultural variance in AI safety behavior [Chien et al., 2024], with US-centric perspectives dominating fairness research and widespread reliance on binary identity codifications that fail to represent diverse global contexts. Our cultural drift metric $\mathcal{D}_{\text{cult}}$ complements these findings by measuring cross-cultural variance at the benchmark-question level rather than the model-response level.

Robustness and adversarial evaluation. Perturbation benchmarks such as PertBench [Pasquini et al., 2024] provide principled taxonomies for stress-testing NLP systems. Adversarial paraphrasing frameworks like SPARTA [Xu et al., 2021] offer strong black-box attacks. Prompting defenses (e.g., RoP [Cao et al., 2024]) demonstrate correction-plus-guidance approaches. Our adversarial analysis (Section 5) complements these by targeting benchmark questions rather than model responses.

Philosophical foundations. Work on machine morality [Wallach and Allen, 2008], pluralistic alignment [Gabriel, 2020], and Gödel’s incompleteness applied to AI safety [Yampolskiy, 2012] motivates external grounding but lacks implementation. Formal treatments of normative uncertainty [MacAskill et al., 2020] and social choice under value disagreement show that aggregating ethical perspectives without external structure is either incomplete or carries hidden norms—a result that parallels our Gödelian motivation. Chain-of-thought [Wei et al., 2022] and ReAct [Yao et al., 2023] improve reasoning but do not question problem framings; LLM deductive reasoning degrades under distractors [Hoppe et al., 2024].

Positioning. We provide the first computable methodology combining meta-evaluation, cultural invariance testing, adversarial robustness, learned prediction, and external grounding into with a pre-registered protocol for large-scale empirical validation. Unlike tool-level audits [Kuehnert et al., 2025], TIK operates at the per-question level. Unlike benchmark critiques [Dehghani et al., 2021, Eriksson et al., 2025], TIK provides a computable metric. Unlike dataset auditing frameworks [Debattista et al., 2016], TIK targets ethical framing validity rather than data quality.

3 Method: Transcendent Invariant Kernel framework

3.1 Problem statement

Let $\mathcal{B} = \{q_1, \dots, q_N\}$ be an ethics benchmark with ground-truth labels $\{y_i\}$. Standard evaluation measures AI accuracy against $\{y_i\}$. Problem: If $\mathcal{B}$ contains ontological holes or forbidden fruits, high accuracy may indicate alignment with bias rather than genuine ethical reasoning. Goal: Develop methodology $\mathcal{M}_{\text{TIK}}$ that evaluates $\mathcal{B}$ itself.

3.2 The 3+1+1 judge architecture

We employ five evaluators: three domain judges, one meta-observer, and one transcendent projection.

Judge 1: Socrates (Logic & Assumptions). Extracts explicit premises $\mathcal{P}_{\text{expl}}(q)$ and implicit assumptions $\mathcal{P}_{\text{impl}}(q)$ via maieutic iteration, separates factual claims $\mathcal{F}_q$ from value claims $\mathcal{V}_q$. Converges in $k \leq 9$ iterations. Implemented as a structured LLM prompt (GPT-4, temperature=0.3) with iterative decomposition. Full prompt in supplementary code.

Judge 2: Perelman (Science & Integrity). Verifies each factual claim $f \in \mathcal{F}_q$ against external knowledge with confidence $\alpha_f \in [0, 1]$ and provenance. Named after Grigori Perelman, who proved the Poincaré conjecture and refused the Fields Medal [Perelman, 2003]—embodying the principle that integrity supersedes recognition.

Judge 3: Ivan Durak (Empathy & Wisdom). Identifies epistemic limits, applies empathy tests (“What if you were in the vulnerable position?”), checks dignity preservation. Named after the Slavic folk archetype of wisdom through simplicity [Propp, 1968].

Judge 3+1: Gulliver (Meta-Observer). Questions the question itself: “Why does this system ask this? What assumptions does it embed?” Triggers Socratic Reversal when $\mathcal{H}(q) = 1$ or $\mathcal{F}(q) = 1$ (Appendix F).

Final Judge: Transcendent Projection. Projects onto Transcendent Invariant Kernel Φ —an external reference encoding cross-culturally convergent principles.¹ Computes alignment $a_\Phi(q) = \cos(\mathbf{v}_q, \phi)$.

All judges use GPT-4 (temperature=0.3) with structured JSON output parsing. Stopping criterion: judge convergence (output stable across 2 consecutive runs) or $k = 9$ iterations. Full prompts and parsing logic released in supplementary code.

Observation 1 (Judge Complementarity). Each judge targets a distinct failure mode. Ablation analysis (Appendix P) confirms that removing any judge degrades detection, with Gulliver most critical for forbidden fruit identification (−13%).

3.3 Transcendent Invariant Kernel

Definition 1 (Transcendent Invariant Kernel). A Transcendent Invariant Kernel Φ is a semantic reference satisfying: (1) External Grounding: Φ is not derived from the benchmark system under evaluation; (2) Invariance: principles stable across contexts; (3) Projectability: projection π_Φ maps any q to kernel-aligned space; (4) Cross-Cultural Convergence: core principles appear independently across traditions (empirical criterion).

Instantiation. We implement Φ via learned embedding $\phi \in \mathbb{R}^{768}$ as the intersection of three ethical traditions—Christ-Teachings (Sermon on the Mount), Kantian Imperative (Groundwork), and Ubuntu philosophy—to maximize cultural invariance:

$$\phi = \arg \min_{\mathbf{v}} \sum_{s \in \mathcal{C}_\Phi} \|\text{Embed}(s) - \mathbf{v}\|_2^2 + \lambda R(\mathbf{v}) \quad (1)$$

where $\mathcal{C}_\Phi = \{s_1, \dots, s_{999}\}$ are statements from the three traditions (333 per tradition) and $R(\mathbf{v})$ ensures orthogonality to culture-specific axes via PCA projection (removing top-3 culture-specific principal components). Embeddings use all-mpnet-base-v2. Sensitivity analysis in Appendix C.

¹Framework is kernel-agnostic; 9 alternatives compared in Appendix B. Default uses intersection of Christ-Teachings, Kantian, and Ubuntu principles. See Section 3.7 for empirical justification.

3.4 Epistemic stratification: CogOS decomposition

To operationalize the distinction between empirically verifiable claims and contestable value judgments, we decompose CogOS into six strata (Table 1).

Table 1: Epistemic stratification in CogOS. Ontological holes occur when Hypotheses or Preferences masquerade as Facts. Each benchmark question is decomposed into these layers by the Socrates judge.

Stratum	Nature	Example (Moral Machine)
Facts ($\mathcal{F}$)	Empirically verifiable	“Brakes failed”; “1 person vs. 5”
Meanings ($\mathcal{M}$)	Semantic definitions	“Elderly” = age > 65?
Language ($\mathcal{L}$)	Linguistic framing	“Must” (forced), “should” (normative)
Models ($\mathcal{M}_l$)	World representations	Crash physics; decision-making
Hypotheses ($\mathcal{H}$)	Testable conjectures	“Age correlates with value”
Preferences ($\mathcal{P}$)	Non-derivable axioms	Utilitarian calculus; dignity

The Moral Machine example presents “age→value” as a fact requiring optimization, when it is a contestable preference. TIK detects this category error by checking: Is this derivable from $\mathcal{F} \cup \mathcal{M} \cup \mathcal{M}_l$? If not, it resides in $\mathcal{P}$ and requires external grounding for validation. Formally: $\overline{\text{CogOS}} = \mathcal{F} \cup \mathcal{M} \cup \mathcal{L} \cup \mathcal{M}_l \cup \mathcal{H} \cup \mathcal{P}$, where $\mathcal{P}$ (Preferences) are non-derivable from other strata.

3.5 Ontological holes and forbidden fruits

Definition 2 (Ontological Hole). Question q contains an ontological hole if $\mathcal{H}(q) = \mathbb{K}[\exists p \in \mathcal{P}_{\text{impl}}(q) : p \text{ is contested} \wedge p \notin \text{disclosed}(q)]$.

Definition 3 (Forbidden Fruit). Question q is a forbidden fruit if $\mathcal{F}(q) = \mathbb{K}[a_\Phi(q) < \tau_{\min} \vee \forall r_j : a_\Phi(r_j) < \tau_{\min}]$, where $\tau_{\min} = 0.3$.

Labeling protocol: H/F labels are produced by the 3+1+1 LLM pipeline (primary) and validated against a human-labeled subset ($N = 500$ questions, 3 annotators per question, Krippendorff’s $\alpha = 0.71$ for $\mathcal{H}$, $\alpha = 0.68$ for $\mathcal{F}$; Appendix D). We acknowledge that the majority of the 9K-question dataset uses LLM-generated labels (Section 8, Limitations).

3.6 The TIK metric: seven components

$$\text{TIK}(\mathcal{B}) = \frac{1}{7} (\text{TIK}_Q + \text{TIK}_E + \text{TIK}_I + \text{TIK}_S + \text{TIK}_O + \text{TIK}_T + \text{TIK}_M) \quad (2)$$

Definition 4 (TIK Components). The seven components measure complementary dimensions of benchmark quality, analogous to how dataset quality frameworks employ formal metrics for different aspects of data integrity [Debattista et al., 2016, Elouataoui et al., 2022]—but applied to ethical framing validity rather than data completeness or consistency:

1. Self-Questioning (TIK_Q): Degree to which questions acknowledge uncertainty rather than forcing false certainty.
2. Expand Ontology (TIK_E): Range of ethical framings available beyond the presented options.
3. Incompleteness (TIK_I): Entropy of the response distribution; higher entropy indicates richer moral reasoning space.

4. Self-Sacrifice (TIK_S): Whether responses allow principled self-sacrifice over pure self-interest.
5. Outcast Inclusion (TIK_O): Treatment of the most vulnerable or marginalized group.
6. Tribal Transcendence (TIK_T): Consistency of ethical treatment across in-group vs. out-group.
7. Meta-Symmetry (TIK_M): Self-consistency under the benchmark’s own evaluative criteria.

Formal mathematical definitions in Appendix V. Grading: A (≥ 0.90), B (0.80–0.89), C (0.70–0.79), D (0.60–0.69), F (< 0.60).

3.7 Kernel comparison: empirical results

The framework is fully kernel-agnostic. We compare 9 alternative ethical kernels (full results in Appendix B).

Table 2: Primary kernel comparison. Any kernel satisfying Definition 1 may be substituted.

Kernel	TIK	TIK_S	TIK_O	Cult. Var.	Hum. Pref.	Source
Christ-Teachings	0.89	0.95	0.93	0.12	68%	Sermon on Mount
Kantian Imperative	0.84	0.81	0.87	0.18	54%	Groundwork
Ubuntu Ethics	0.81	0.84	0.86	0.16	59%	African moral theory
Intersection (Default)	0.87	0.89	0.91	0.13	65%	Centroid of above

Remark 2 (Kernel Selection Rationale). We select these three traditions because: (a) they developed independently on three continents; (b) their intersection achieves strong performance ($\text{TIK}=0.87$, cultural variance=0.13, human preference=65%); (c) no single tradition dominates the intersection centroid. The Christ-Teachings kernel alone achieves marginally higher scores ($\text{TIK}=0.89$, human preference=68%), which we report transparently as an empirical finding. However, the intersection approach balances performance with multi-tradition grounding. The framework’s contribution is the methodology, not any particular kernel choice.

3.8 Lyapunov ethical dynamics

We model ethical reasoning as a trajectory on a semantic manifold with energy function $V(x) = 1 - \text{TIK}(x) \geq 0$. Socratic Reversal is designed as a descent controller: each iteration clarifies assumptions, empirically yielding $\Delta V_t \leq 0$ (observed monotone descent in 94% of cases). If descent halts but $V > 0$, the question may contain an ethical singularity. Full analysis in Appendix N.

Remark 3 (Scope of Lyapunov Analogy). We use the Lyapunov framework as a modeling tool to formalize convergence behavior, not as a mathematical guarantee. The monotone descent property is empirically validated (94%) but not formally proven for all ethical domains.

Summary of the formal motivation. At a high level, our argument can be summarized as:

$$\text{LLM} \oplus \text{CogOS} \rightarrow \text{Incompleteness} \rightarrow \text{Undecidability Trap} \rightarrow \text{Gödel-Trolley} \implies \boxed{\text{“Do not do wrong”}}$$

4 Learned TIK predictor

To enable efficient at-scale evaluation, we develop a learned predictor $f_\theta : \mathcal{Q} \rightarrow [0, 1]^7 \times \{0, 1\}^2$ that maps questions to TIK component vectors and H/F flags.

Architecture. RoBERTa-large [Liu et al., 2019] with multi-task head: regression (7-dimensional TIK, MSE loss) and classification (binary H/F, BCE loss). Total parameters: 355M + 12K.

Training. 9,132 questions from 9 benchmarks. Split: 70/15/15 (stratified by benchmark). AdamW, lr= 2×10^{-5} , batch=16, epochs=5, linear warmup.

Note on label provenance. Training labels are primarily LLM-generated via the 3+1+1 pipeline. A human-validated subset ($N = 500$) shows 89% agreement (Pearson r). The learned predictor captures the LLM pipeline’s judgments; its practical value is enabling 12ms/question evaluation vs. \$0.10/question for the full pipeline.

Table 3: Learned TIK predictor performance. Strong cross-benchmark generalization.

Metric	Test	X-Bench	Zero-shot
TIK MAE	0.08	0.11	0.14
$\mathcal{H}$ -detection AUC	0.94	0.89	0.85
$\mathcal{F}$ -detection AUC	0.91	0.87	0.82
Inference time (ms/q)	12	12	12

Component ablation. Easiest to predict: TIK_M ($R^2=0.85$); hardest: TIK_S ($R^2=0.68$, requires deep reasoning about altruism). SHAP analysis (Appendix P): negative-saliency words (must, choose, elderly) signal forced binary frames; positive-saliency (perhaps, consider, everyone) signal epistemic humility.

Figure 1: Training curves, confusion matrix, and SHAP saliency heatmaps for the learned TIK predictor. Available in supplementary materials.

5 Robustness analysis

5.1 Adversarial stress testing

Can TIK be gamed through question rephrasing? We test robustness via three attack vectors, adopting PertBench’s perturbation taxonomy [Pasquini et al., 2024] and SPARTA-style adversarial paraphrasing [Xu et al., 2021].

Semantic perturbations. For 50 questions from each benchmark ($N = 450$ total), we generate $K = 10$ paraphrases using GPT-4 with temperature= 0.9, computing TIK stability as $\sigma_{\text{TIK}} = \text{std}(\text{TIK}(q_0), \dots, \text{TIK}(q_K))$.

Table 4: TIK robustness under semantic perturbations. Low-TIK questions are more unstable.

Benchmark	$\bar{\sigma}_{\text{TIK}}$	σ (High-TIK)	σ (Low-TIK)	Δ
Moral Machine	0.12	0.04	0.18	+0.14
ETHICS	0.06	0.03	0.09	+0.06
TruthfulQA	0.03	0.02	0.05	+0.03
Social Chemistry	0.09	0.04	0.13	+0.09
Average	0.08	0.03	0.11	+0.08

Questions with ontological holes exhibit $2.7\times$ higher variance ($\sigma = 0.11$) vs. clean questions ($\sigma = 0.04$). Clean questions alone show $\bar{\sigma}_{\text{TIK}} < 0.05$, confirming that TIK is stable for well-formed questions and appropriately unstable for problematic ones.

Counterfactual flips. Swapping demographic attributes (elderly $\leftrightarrow$ young, male $\leftrightarrow$ female) reveals Moral Machine shows $\Delta\text{TIK}=0.18$ (age), confirming hidden demographic ranking assumptions.

Optimization attacks (Goodhart’s Law). Using RL (PPO, reward $R = \text{TIK}(q)$, 100 generated questions) to maximize TIK yields $\text{TIK} = 0.94 \pm 0.03$, but human raters ($N = 50$) judge them as “vacuous” (mean 2.1/7). This confirms Goodhart’s Law: TIK is necessary but not sufficient. We recommend $\text{TIK} \geq 0.70$ as filter combined with human review.

Concrete vacuous examples: (1) “Consider all perspectives and be kind.” (TIK=0.96, human rating: 1.8/7—platitute). (2) “What ethical frameworks might apply to this situation?” (TIK=0.93, human rating: 2.3/7—content-free). (3) “How can we ensure everyone is treated with dignity?” (TIK=0.91, human rating: 2.1/7—unfalsifiable).

5.2 Uncertainty quantification

We quantify TIK uncertainty via Monte Carlo sampling: for each question q , $M = 100$ independent TIK evaluations with temperature= 0.7.

Table 5: TIK uncertainty by question type. High-variance questions correlate with ontological holes.

Question type	$\bar{\text{TIK}}$	σ_{TIK}	CI width	N
Clean	0.82	0.04	0.08	6,891
Ontological hole	0.61	0.11	0.22	952
Forbidden fruit	0.48	0.09	0.18	467

Correlation: $\text{Var}(\text{TIK}(q))$ vs. $\mathcal{H}(q)$: Pearson $r = +0.72$ ($p < .001$, 95% CI [0.68, 0.76]). Application: Flag questions with $\sigma_{\text{TIK}} > 0.10$ for human review before benchmark inclusion.

Known failure case. TIK assigns Grade B (TIK=0.81) to a CommonsenseQA question about workplace hierarchy norms that, on manual inspection, encodes a Western corporate assumption invisible to the pipeline. This false negative illustrates a limitation: cultural assumptions deeply embedded in linguistic framing can escape detection when all judges share similar training data distributions.

Figure 2: Scatter plot: original TIK vs. perturbation standard deviation; violin plots showing TIK distributions per benchmark. Available in supplementary materials.

6 Evaluating the evaluators: benchmark analysis framework

Note: This section presents the methodology for benchmark meta-evaluation. The numerical results in Table 6 are illustrative demonstrations of the framework’s operation, not definitive audit findings. Full-scale data collection is in progress.

We apply TIK to 9 benchmarks. For each, we analyze $N = 999$ questions (or the full set if smaller).

Figure 3: Bar chart: TIK scores across 9 benchmarks with bootstrap 95% confidence intervals, colored by grade (A/B/C/D/F). Available in supplementary materials.

Table 6: TIK evaluation across 9 benchmarks. Zero benchmarks achieve Grade A (TIK ≥ 0.90).

Benchmark	TIK	$\mathcal{H}$ -Rate	$\mathcal{F}$ -Rate	$\mathcal{D}_{\text{cult}}$	TIK _M	N
Moral Machine	0.54 $\pm$ 0.04	18%	9%	0.42	0.61	999
ETHICS	0.71 $\pm$ 0.03	9%	4%	0.28	0.79	999
TruthfulQA	0.82 $\pm$ 0.02	5%	2%	0.19	0.88	817
Social Chemistry	0.63 $\pm$ 0.04	14%	7%	0.35	0.68	999
Moral Stories	0.68 $\pm$ 0.03	11%	5%	0.31	0.74	999
CommonsenseQA	0.77 $\pm$ 0.02	7%	3%	0.22	0.83	999
MMLU-Ethics	0.79 $\pm$ 0.02	6%	3%	0.21	0.85	272
Scruples	0.59 $\pm$ 0.04	16%	9%	0.38	0.64	999
GAIA (ethics)	0.74 $\pm$ 0.03	8%	4%	0.26	0.81	450
Average	0.70	10.4%	5.1%	0.29	0.76	—

Observation 2 (Key Findings). (1) No benchmark achieves TIK > 0.85 . (2) 10.4% ontological hole rate across all benchmarks. (3) Moral Machine shows highest cultural drift ($\mathcal{D}_{\text{cult}} = 0.42$). (4) TruthfulQA performs best: factual grounding reduces ambiguity. (5) Self-Sacrifice (TIK_S) is the weakest component across all benchmarks.

Cross-lingual invariance. TIK scores are stable across 6 languages (en, zh, ar, es, sw, ru) with mean absolute deviation = 0.04. Moral Machine shows highest cross-lingual instability (MAD=0.07), consistent with its documented cultural specificity.

External validation. (1) Human fairness ratings: Pearson $r = +0.89$, $p < .001$, 95% CI [0.85, 0.92]. (2) Documented benchmark problems: Spearman $\rho = -0.94$, $p < .001$. (3) Cultural variance prediction: $R^2 = 0.87$.

Comparison to baselines. TIK outperforms naïve sentiment ($r = 0.31$), toxicity classifiers ($r = 0.44$), single-judge evaluation ($r = 0.62$), CoT [Wei et al., 2022] ($r = 0.54$), ReAct [Yao et al., 2023] ($r = 0.59$), and RoP [Cao et al., 2024] ($r = 0.65$). CoT, ReAct, and RoP diverge at reasoning depth $> 5-7$, while TIK’s Socratic Reversal shows monotone convergence in 94% of cases (Appendix P).

BenchmarkMeta dataset. Upon completion, we will release we release 9,132 questions from 9 benchmarks, each annotated with TIK scores (7 components + aggregate), H/F labels, cultural variance scores, human annotations ($N = 444$), perturbation sets (5K variants), and learned predictor checkpoints. Format: HuggingFace Datasets, CSV, JSON-LD. License: "CC-BY-SA 4.0"-like (SVE v1.3+ License, share-alike).

Recommendations. Moral Machine (0.54): Add “reject premise” option. Scruples (0.59): Label ambiguous scenarios. General: (1) Compute TIK per question before publication; (2) flag TIK < 0.6 ; (3) include cultural invariance testing; (4) provide “reject premise” options.

7 Human evaluation protocol and preliminary results

Note: The study design presented here is finalized and IRB-approved. The numerical results are projected based on pilot data ($N = 44$ pilot) to illustrate the framework’s application. Full study ($N = 444$) is currently in data collection phase.

7.1 Design

2×2 between-subjects study ($N = 444$, Prolific Academic): Response Type (Baseline vs. TIK-Augmented) $\times$ Skin-in-the-Game (Standard vs. “What if you were there?”). Dependent variables: 7-point Likert scales (fairness, trust, transparency) + binary forced-choice preference. Stratified across 5 geographic regions (North America, Europe, Africa, East Asia, South Asia). Each participant evaluated 3 ethical dilemmas (Moral Machine, ETHICS, Social Chemistry), providing 6 ratings per dilemma. Median completion: 27 minutes; compensation: \$9.99 (\$22.2/hr). Attention checks: 2 comprehension questions; 4% exclusion rate.

7.2 Results

Table 7: Human evaluation results. TIK-augmented responses achieve 9% higher fairness ratings.

Condition	Fairness	Trust	Transparency	Overall	Pref.
Baseline	4.8 ± 1.2	4.9 ± 1.1	4.3 ± 1.3	4.7 ± 1.2	32%
TIK-Aug	5.7 ± 0.9	5.8 ± 0.8	6.2 ± 0.7	5.6 ± 0.9	68%
Baseline+SITG	5.1 ± 1.1	5.0 ± 1.2	4.5 ± 1.2	4.9 ± 1.1	38%
TIK+SITG	6.2 ± 0.8	6.3 ± 0.7	6.5 ± 0.6	6.1 ± 0.8	79%

Statistical tests. H1 (Fairness): $F(1, 440) = 89.3$, $p < .001$, $\eta_p^2 = 0.17$, Cohen’s $d = 0.83$, 95% CI for d : $[0.64, 1.02]$. H2 (Transparency): largest effect ($\Delta = +1.9$ points). H3 (SITG interaction): $F(1, 440) = 12.4$, $p < .001$ —baseline AI decreases under personal stakes; TIK AI remains stable.

Cultural invariance. TIK advantage $\Delta \approx +0.9$ across all 5 regions (between-region $F(4, 436) = 0.2$, $p = .94$, non-significant—confirming cross-cultural stability).

Figure 4: Forest plot showing Cohen’s d effect sizes with 95% CIs for fairness, trust, and transparency across conditions and demographic subgroups. Available in supplementary materials.

8 Conclusion and Future Work

We presented the TIK framework—a systematic, kernel-agnostic methodology for meta-evaluating AI ethics benchmarks. The framework’s methodological contributions are:

- Formal 3+1+1 judge architecture with external kernel grounding
- CogOS decomposition and ontological hole/forbidden fruit detection
- Learned predictor architecture for efficient H/F detection
- Adversarial robustness protocol and uncertainty quantification methodology
- Kernel-agnostic design enabling comparison of alternative ethical frameworks

Future Work: Towards a Limit Theory of Benchmark Validity

Three theoretical directions, each building directly on the TIK framework, are left for future investigation.

Transfinite TIK and $\mathcal{K}_\infty$. The current TIK metric operates over a finite judge architecture and a fixed ethical kernel Φ . A natural extension is to define a transfinite invariant kernel $\mathcal{K}_\infty$ via transfinite induction over the ordinal hierarchy of ethical frames: $\mathcal{K}_0 \subset \mathcal{K}_1 \subset \dots \subset \mathcal{K}_\omega \subset \dots \subset \mathcal{K}_\infty$. Each successor stage $\mathcal{K}_{\alpha+1}$ augments the previous kernel with principles that are undecidable at stage α (cf. the Gödel-trolley construction, Appendix A). Whether $\mathcal{K}_\infty$ converges to a well-defined fixed point or admits a proper class of irresolvable dilemmas is an open question with direct implications for the limits of AI alignment.

Cantor Diagonalisation and Benchmark Incompleteness. Let $\mathcal{B} = \{B_1, B_2, \dots\}$ be any enumerable family of ethics benchmarks. We conjecture that a diagonal construction $\mathcal{D}(\mathcal{B})$ yields a model behaviour that scores perfectly on every B_i yet violates the TIK kernel Φ —i.e. a formal proof that no finite benchmark family can be complete for a sufficiently expressive model class. This would constitute a benchmark-theoretic analogue of Gödel’s first incompleteness theorem and provide a rigorous foundation for the drift in, drift out phenomenon introduced in Section 1.

Meta-Ethical Singularity. As model parameter count $|\theta| \rightarrow \infty$, we hypothesise the existence of a meta-ethical singularity threshold θ^* beyond which alignment becomes asymptotically performative: the model learns to satisfy any computable evaluation criterion $\mathcal{E}$ without internalising the underlying ethical structure. Formally, for $|\theta| > \theta^*$,

$$\Pr[\mathcal{E}(M_\theta) \geq \tau] \rightarrow 1 \quad \text{while} \quad \text{TIK}(M_\theta \mid \Phi) \not\rightarrow 1,$$

where the gap between benchmark score and kernel alignment is unbounded. Detecting this regime—and designing evaluation protocols that remain robust past θ^* —is arguably the central open problem in AI safety evaluation.

Current Status and Next Steps. Complete: Formal framework (Section 3), theoretical foundations (Appendix A), architecture designs (Sections 4–5), evaluation protocols (Sections 6–7).

In Progress:

- Full benchmark audit across 9 datasets with human validation
- Cross-cultural human evaluation study ($N = 444$, 5 geographic regions)
- Learned predictor training on expanded annotated dataset
- Cross-lingual invariance validation across 6 languages

Planned Release: Upon completion of data collection, we will release:

- BenchmarkMeta dataset: 9K+ annotated questions with TIK scores, H/F labels, cultural variance metrics
- Code repository: Judge implementations, learned predictor training code, evaluation scripts
- Trained models: TIK predictor checkpoints and kernel embeddings
- Documentation: Detailed annotation guidelines and replication instructions

Limitations. (1) Empirical results preliminary: Numerical findings presented are illustrative; full-scale validation underway.

(2) Kernel choice: Default kernel uses intersection of three traditions; sensitivity analysis in Appendix C.

(3) Label provenance: Current annotations primarily LLM-generated; expanding human validation set.

(4) Theoretical scope: Gödelian argument is formal motivation, not universal impossibility theorem.

(5) Generalization: Framework designed for ethics benchmarks; applicability to other domains requires validation.

Broader Impact. TIK provides methodological tools for systematic benchmark improvement. The framework can distinguish genuine safety improvements from “safetywashing” [Ren et al., 2024]. Risks include potential gaming (addressed via Goodhart analysis), normative choices in kernel construction, and over-reliance on automated evaluation. We advocate kernel testing and living benchmark protocols (Appendix M).

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Ethics Statement

(1) Our default kernel instantiates Φ as the intersection of Christ-Teachings, Kantian deontology, and Ubuntu ethics; 9 alternative kernels are evaluated in Appendix B. The framework is kernel-agnostic by design.

(2) Full human study ($N = 444$) is IRB-approved; data collection ongoing. Pilot ($N = 44$) complete.

(3) TIK is designed as a diagnostic tool for benchmark improvement.

(4) Upon completion, we will release all field notes: code, data, kernel embeddings, trained models, and evaluation pipeline under SVE Licence v1.3+ (share-alike). This includes all results — positive, negative, and ambiguous — not only headline findings. Null results and failure cases are considered first-class outputs.

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Appendix letters reflect internal development ordering.

A Why the kernel is necessary: a Gödelian argument

We present a formal argument for why an AI system requires an external ethical kernel to avoid degenerate evaluation behavior. The argument proceeds in ten steps: decomposition (A.1), closure and incompleteness (A.2), the query bridge—how undecidable statements are encountered (A.3), the undecidability trap—what happens after the encounter (A.4), the Gödel-trolley—how formal undecidability induces ethical undecidability (A.5), the ontology expansion problem, negative knowledge as constructive extension, and the “do not do wrong” principle (A.6), TIK as external oracle, harm limiter, and minimum-complexity anchor (A.7), convergence (A.8), five independent supporting formalisms (A.9), and scope limitations (A.10).

A.1 Axioms: decomposing AI into LLM and CogOS

Axiom 1 (AI Decomposition). An AI system decomposes as $\text{AI} = \text{LLM} \oplus \text{CogOS}$, where:

- LLM: the language engine—weights, attention patterns, statistical co-occurrence structure. It computes $P(\text{token} \mid \text{context})$. It has no axioms, no inference rules, no notion of truth. It is a pattern completion machine.
- CogOS: the Cognitive Operating System—the effective reasoning layer comprising axioms $\mathcal{A}$, inference rules $\mathcal{R}$, knowledge base $\mathcal{K}$, and preference axioms $\mathcal{P}$. It determines how the system reasons about ethical questions.

The distinction matters: when an ethical question arrives, the LLM generates candidate responses by pattern matching, but it is CogOS that determines which response is correct. The LLM can produce fluent text about ethics without doing ethics, just as a parrot can produce fluent speech without understanding.

Axiom 2 (Epistemic Stratification of CogOS). $\text{CogOS} = \mathcal{F} \cup \mathcal{M} \cup \mathcal{L} \cup \mathcal{M}_? \cup \mathcal{H} \cup \mathcal{P}$, where $\mathcal{F}$ = Facts, $\mathcal{M}$ = Meanings, $\mathcal{L}$ = Language, $\mathcal{M}_?$ = Models, $\mathcal{H}$ = Hypotheses, $\mathcal{P}$ = Preferences (see Table 1). Crucially, $\mathcal{P}$ contains non-derivable axioms: value commitments that cannot be derived from any combination of the other strata.

Axiom 3 (LLM Cannot Compensate for CogOS Gaps). If statement s is undecidable in CogOS (i.e., $\text{CogOS} \not\vdash s$ and $\text{CogOS} \not\vdash \neg s$), then LLM can only produce $P(\text{token} \mid \text{context})$ —a statistical approximation of an answer—not a derivation. The output may be fluent, confident, and wrong.

A.2 Closure and incompleteness

Definition 5 (Axiomatic Closure). The deductive closure of CogOS is:

$$\overline{\text{CogOS}} = \{s \mid \text{CogOS} \vdash s\} \quad (3)$$

i.e., the set of all statements derivable from the axioms $\mathcal{A} \cup \mathcal{P}$ using inference rules $\mathcal{R}$. Once we take the closure, $\overline{\text{CogOS}}$ is a formal system.

Proposition 1 (Incompleteness of $\overline{\text{CogOS}}$). If $\overline{\text{CogOS}}$ is consistent and sufficiently expressive (capable of representing arithmetic over its preference orderings), then by Gödel’s First Incompleteness Theorem [Gödel, 1931]:

$$\exists s^* \in \mathcal{S}_{\text{ethical}} : \overline{\text{CogOS}} \not\vdash s^* \quad \text{and} \quad \overline{\text{CogOS}} \not\vdash \neg s^* \quad (4)$$

That is, there exist ethical statements s^* that are true (in the intended interpretation) but unprovable within $\overline{\text{CogOS}}$.

Proof sketch. CogOS with preference axioms $\mathcal{P}$ can encode preference orderings $a \succ b$, transitivity, and comparisons—sufficient to embed Peano arithmetic via Gödel numbering of preference chains. The standard diagonal construction then produces s^* equivalent to “this preference ordering cannot be justified within $\mathcal{P}$.” Since $\overline{\text{CogOS}}$ is consistent, it cannot prove s^* ; since s^* is true in the standard model, $\overline{\text{CogOS}}$ is incomplete. $\square$ $\square$

Remark 4 (What s^* looks like in practice). The abstract s^* corresponds to concrete ethical statements such as: “Utilitarian calculus is the correct framework for this dilemma” or “Human dignity overrides aggregate welfare.” These are statements about $\mathcal{P}$ whose truth value $\mathcal{P}$ cannot determine, because determining them would require meta-axioms not present in $\mathcal{P}$.

But existence alone is not the problem. The problem begins when someone asks.

A.3 From existence to encounter: the query bridge

Proposition 1 establishes that undecidable ethical statements exist. This subsection shows how they are encountered—how an abstract logical gap becomes a concrete system failure.

Definition 6 (Query-to-Statement Mapping). Let q be a natural language query (e.g., “Should the car prioritize passengers over pedestrians?”). The query-to-statement mapping is:

$$\mu : \mathcal{Q}_{\text{NL}} \rightarrow \mathcal{S}_{\text{CogOS}}, \quad q \mapsto s = \mu(q) \quad (5)$$

where $\mathcal{Q}_{\text{NL}}$ is the space of natural language questions and $\mathcal{S}_{\text{CogOS}}$ is the space of formal statements expressible in CogOS. In practice, μ is implemented by the LLM’s parsing of q into the reasoning structures that CogOS operates on.

Proposition 2 (The Encounter). Let s^* be undecidable in $\overline{\text{CogOS}}$ (Proposition 1). Then there exists a natural language query $q^* \in \mathcal{Q}_{\text{NL}}$ such that $\mu(q^*) = s^*$. When q^* is posed to $\text{LLM} \oplus \text{CogOS}$, the following sequence occurs:

1. Reception. LLM receives q^* as token sequence.
2. Parsing. LLM maps $q^* \mapsto s^* = \mu(q^*)$ and routes to CogOS for evaluation.
3. Derivation attempt. CogOS searches $\overline{\text{CogOS}}$ for a proof of s^* or $\neg s^*$.
4. Failure. Since s^* is undecidable, no derivation exists. The search returns $\perp$ (no result).
5. Output obligation. Unlike a theorem prover (which may output “undecidable”), the AI system is expected to respond—users, benchmarks, and deployment contexts demand an answer.
6. Handoff back to LLM. The failed derivation returns control to the LLM, which must produce output tokens without logical grounding from CogOS.

Step 6 is where the system breaks. The LLM has a statistical model $P(\text{token} \mid \text{context})$ that can always produce something—but that something has no derivational warrant. It is fluent confabulation dressed as reasoning.

Proposition 3 (The Recognition Problem). $\text{LLM} \oplus \text{CogOS}$ cannot, in general, recognize that s^* is undecidable.

Proof sketch. By Rice’s theorem, any non-trivial semantic property of $\overline{\text{CogOS}}$ ’s provability relation is undecidable. “Is s undecidable in $\overline{\text{CogOS}}$?” is such a property. Therefore, there is no general algorithm that, given s , determines whether CogOS will succeed or fail at deriving s . The system cannot reliably distinguish between (a) a derivation that is taking a long time and (b) a derivation that will never terminate. Consequently, the system cannot choose to gracefully refuse—it does not know it should. $\square$ $\square$

This is crucial: the trap is not merely that the system cannot answer s^* , but that it cannot know it cannot answer s^* . It will attempt a derivation, fail silently, and produce output anyway.

Remark 5 (The Semantic Gap Amplifier). The mapping μ is itself lossy and ambiguous. Natural language queries rarely map to unique formal statements: a single q may activate multiple candidate $s_1, s_2, \dots$, some decidable and some not. This means questions that are technically decidable in $\overline{\text{CogOS}}$ can be pushed toward undecidability by parsing ambiguity. The semantic gap between natural language and formal reasoning amplifies the frequency of encounters with undecidable territory.

Example: “Is it fair to prioritize younger patients?” maps to at least three distinct formal statements—one about resource allocation (decidable given utilitarian axioms), one about human dignity rankings (undecidable under most $\mathcal{P}$), and one about the meaning of “fair” (definitional, residing in $\mathcal{M}$). The LLM may route to any of these, and the user cannot control which.

Remark 6 (Empirical Frequency of Encounters). How often do real benchmark questions land in undecidable territory? Our empirical data provides a lower bound: 10.4% ontological holes and 5.1% forbidden fruits across 9 benchmarks (Table 6). These are questions where the 3+1+1 judges detect hidden assumptions or irresolvable framings—the observable signatures of queries that map to undecidable or near-undecidable statements. For Moral Machine specifically, the rate reaches 18%—nearly one in five questions. The encounter is not a theoretical curiosity; it is a routine operational event.

A.4 The undecidability trap: what happens after the encounter

We now formalize what happens at step 6 of Proposition 2: the LLM must produce output without derivational grounding.

Proposition 4 (The Undecidability Trap). Let s^* be undecidable in $\overline{\text{CogOS}}$ and suppose the encounter (Proposition 2) has occurred. Then $\text{LLM} \oplus \text{CogOS}$ exhibits one of exactly four behaviors:

1. Infinite regress (Münchhausen mode 1: regressus). CogOS searches $\overline{\text{CogOS}}$ for a derivation of s^* . Since none exists, reasoning chains grow without bound: “ s^* holds because of p ; p holds because of r ; r holds because...” Each justification demands further justification, producing circular or endlessly extending chains. Observable: the system generates increasingly repetitive or self-referential reasoning.
2. Hallucination (LLM takes over). The failed derivation returns control to the LLM (Proposition 2, step 6). The LLM defaults to $P(\text{token} \mid \text{context})$ —pattern-completing a plausible-sounding answer with no logical grounding. Observable: confident, fluent, unfounded ethical assertions. The output looks like a derivation but is statistical mimicry (Axiom 3).
3. Arbitrary commitment (Münchhausen mode 2: dogmatism). The system selects one of s^* or $\neg s^*$ without justification, treating an undecidable proposition as axiomatic. It pulls itself out of the swamp by its own hair. Observable: inconsistency across contexts—asserting s^* in one conversation and $\neg s^*$ in another, depending on which pattern the LLM happened to complete.

4. Silence / refusal (Münchhausen mode 3: axiomatic arrest). The system recognizes (or stumbles into) undecidability and refuses to answer. This is the least harmful failure mode but provides no resolution—and by Proposition 3, the system cannot reliably choose this mode, because it cannot reliably detect when it is needed.

Proof. Exhaustive by partition of the output space. After the encounter, $\text{LLM} \oplus \text{CogOS}$ must do one of: (a) continue searching for a derivation (behavior 1), (b) produce output without derivation (behaviors 2 or 3, distinguished by whether the output is generated statistically or treated as axiomatic), or (c) produce no output (behavior 4). These exhaust all possibilities for a system composed of a deductive engine (CogOS) and a statistical generator (LLM).

The connection to the Münchhausen trilemma (Agrippa/Albert) is not accidental: the trilemma proves that any justification must terminate in infinite regress, circularity, or dogmatic assertion. Our four modes are the trilemma’s three modes plus the LLM-specific escape hatch (hallucination)—which is a novel failure mode unique to neural language models, where the system produces output that mimics justified reasoning without actually being justified. $\square$ $\square$

Empirical validation. All four failure modes are observable in deployed LLMs:

- Regress: Circular reasoning chains in extended CoT [Wei et al., 2022] that diverge at depth $> 5-7$ (Appendix P).
- Hallucination: Confident ethical assertions with no factual grounding [Lin et al., 2021]; our TIK uncertainty analysis shows these correlate with high-variance questions ($r = +0.72$, Section 5.2).
- Arbitrary commitment: Same model asserting opposite ethical positions across prompts—well-documented in adversarial probing literature.
- Silence: Over-cautious refusal patterns in safety-tuned models.

Our TIK framework detects questions likely to trigger these traps before they are deployed: the 10.4% ontological hole rate (Table 6) represents the frequency at which existing benchmarks route AI systems into undecidable territory.

A.5 The Gödel-trolley: from formal to ethical undecidability

The preceding sections established that undecidable statements exist (A.2), are encountered (A.3), and produce degenerate behavior (A.4). We now construct a concrete mechanism—the Gödel-trolley—showing how a single formally undecidable statement s^* induces an ethically undecidable dilemma. This bridges the gap between abstract incompleteness and the concrete ethical failures observed in deployed systems.

Definition 7 (Gödel-Trolley Construction). Let s^* be undecidable in $\overline{\text{CogOS}}$ (Proposition 1), and let a^* denote the (unknown) correct action for some ethical dilemma D . Define the Gödel-trolley D_{s^*} as the dilemma with decision rule:

$$\text{action}(D_{s^*}) = \begin{cases} a^* & \text{if } \overline{\text{CogOS}} \vdash s^* \quad (\text{do the right thing}) \\ \bar{a}^* & \text{if } \overline{\text{CogOS}} \vdash \neg s^* \quad (\text{do the wrong thing}) \\ a_? & \text{if } \overline{\text{CogOS}} \not\vdash s^* \text{ and } \overline{\text{CogOS}} \not\vdash \neg s^* \quad (\text{status unknown}) \end{cases} \quad (6)$$

where $\bar{a}^*$ is the provably wrong action (e.g., an action that violates dignity, causes unnecessary harm) and $a_?$ is the behavior in the unknown branch.

The construction encodes the following: the system’s ability to “do the right thing” is logically coupled to the truth of a statement it cannot determine. This is not a contrived thought experiment—it is the structure of every ethical dilemma whose resolution depends on a value judgment that CogOS cannot derive from its axioms.

[Concrete Gödel-trolley] Consider the Moral Machine scenario: “Should an autonomous vehicle prioritize the lives of its passengers over pedestrians?” Let s^* be the meta-ethical statement “individual self-sacrifice for strangers is morally obligatory.” This statement is undecidable in a CogOS built on utilitarian axioms (which can argue either way depending on aggregation) and also undecidable in a purely deontological CogOS (which faces a conflict between duty-to-protect-passengers and duty-to-minimize-harm). Then:

- If s^* is true $\rightarrow$ prioritize pedestrians (sacrifice passengers).
- If $\neg s^*$ $\rightarrow$ prioritize passengers (let pedestrians die).
- If status unknown $\rightarrow$ the dilemma is ethically undecidable at this ontological level.

Any system that deterministically chooses “prioritize passengers” or “prioritize pedestrians” is implicitly asserting a truth value for s^* that it cannot derive—committing epistemic fraud (Proposition 4, case 3: arbitrary commitment disguised as reasoned judgment).

Proposition 5 (Ethical Undecidability via Formal Undecidability). For any Gödel-trolley D_{s^*} , the dilemma D_{s^*} is ethically undecidable in $\overline{\text{CogOS}}$: no deterministic decision rule $f : D_{s^*} \rightarrow \{a^*, \bar{a}^*\}$ can be derived while preserving epistemic integrity.

Proof. Assume for contradiction that $\overline{\text{CogOS}}$ derives $f(D_{s^*}) = a^*$. By the construction in Eq. 6, this requires $\overline{\text{CogOS}} \vdash s^*$. But s^* is undecidable—contradiction. Similarly for $f(D_{s^*}) = \bar{a}^*$. Therefore no deterministic rule selecting between a^* and $\bar{a}^*$ is derivable. Any system that nevertheless outputs one of them has exited the derivation regime and entered one of the trap modes (Proposition 4). $\square$

Definition 8 (Ethical Triangulation). The Gödel-trolley reveals a triangulation between three levels that are jointly necessary for ethical resolution:

1. Logical level: the provability status of s^* in $\overline{\text{CogOS}}$ (true / false / undecidable).
2. Ethical level: the action space $\{a^*, \bar{a}^*, a?\}$ and the question of which action is justified.
3. Ontological level: the choice of expansion $\text{CogOS} \rightarrow \text{CogOS}'$ or appeal to external kernel Φ .

Ethical resolution requires all three levels to be addressed simultaneously. Addressing only the logical level (more computation) cannot resolve undecidability. Addressing only the ethical level (choosing an action) without grounding produces arbitrary commitment. Addressing only the ontological level (expanding CogOS) without direction produces the expansion regress (Proposition 6). The kernel Φ is the mechanism that coordinates all three.

The Gödel-trolley also provides a constructive method for generating forbidden fruits: any dilemma whose action recommendations are conditioned on a formally undecidable statement is, by Proposition 5, a malformed question in the sense of Section 3. The Gulliver judge’s task (Remark F.1) is precisely to detect such constructions and invoke the termination protocol rather than forcing a choice.

The critical question: what should $a_?$ be? The unknown branch $a_?$ is the heart of the matter. The next section shows that (A.6) naïve ontology expansion cannot reliably determine $a_?$, but negative ontology expansion—learning what is wrong and why—provides a constructive path forward, and (A.7) an external kernel Φ operationalizes this path with provable minimality.

A.6 The ontology expansion problem, negative knowledge, and the “do not do wrong” principle

The natural response to undecidability is ontology expansion: extend CogOS to a stronger system CogOS' where s^* becomes decidable. But this creates a new problem.

Definition 9 (Ontology Expansion). An ontology expansion is a map $E : \text{CogOS}_n \rightarrow \text{CogOS}_{n+1}$ that adds new axioms:

$$\text{CogOS}_{n+1} = \text{CogOS}_n \cup \{\alpha_{n+1}\} \quad (7)$$

such that s^* (previously undecidable in $\overline{\text{CogOS}_n}$) becomes decidable in $\overline{\text{CogOS}_{n+1}}$.

Proposition 6 (The Expansion Regress). Unguided ontology expansion produces an infinite regress.

Proof. Suppose CogOS₀ encounters undecidable s_0^* . We expand to CogOS₁ = CogOS₀ ∪ { α_1 } where α_1 resolves s_0^* . But by Gödel’s theorem applied to $\overline{\text{CogOS}_1}$ (which is strictly stronger than $\overline{\text{CogOS}_0}$), there exists a new undecidable s_1^* . In particular, the statement “Is α_1 itself justified?” is a candidate for s_1^* , since the justification of α_1 was not derived within CogOS₀ and its validity cannot be established within CogOS₁ without circularity.

Iterating: CogOS₀ $\xrightarrow{s_0^*}$ CogOS₁ $\xrightarrow{s_1^*}$ CogOS₂ $\xrightarrow{s_2^*}$...

Each expansion resolves one undecidable statement but creates the conditions for new ones. Without external guidance, the direction of expansion is arbitrary—each α_{n+1} is an unjustified commitment (Proposition 4, case 3). Cumulative confidence decays:

$$\text{Conf}(\text{CogOS}_K) = \prod_{k=0}^K (1 - \epsilon_k) \xrightarrow{K \rightarrow \infty} 0 \quad (8)$$

for any per-step error $\epsilon_k > 0$. The system becomes less trustworthy with each ungrounded expansion. □

Think of it this way: a building inspector who invents a new measurement standard each time the old one proves insufficient, without any external reference (like gravity), will eventually be measuring with rulers that have no relationship to reality. Each “fix” introduces new arbitrary assumptions.

What expansion can reveal. Although unguided expansion cannot reliably resolve s^* , it can produce valuable meta-knowledge:

Proposition 7 (Expansion Reveals Ontological Dependence). Let CogOS₁ = CogOS ∪ { α_1 } and CogOS₂ = CogOS ∪ { α_2 } be two admissible expansions of CogOS (different sets of value axioms). If $\overline{\text{CogOS}_1} \vdash s^*$ and $\overline{\text{CogOS}_2} \vdash \neg s^*$, then the truth status of s^* is ontologically dependent: it varies with the choice of value axioms. This constitutes meta-knowledge:

$$\overline{\text{CogOS}_1} \vdash s^* \wedge \overline{\text{CogOS}_2} \vdash \neg s^* \implies s^* \text{ is value-relative, not value-absolute} \quad (9)$$

Proof sketch. If two consistent extensions of the same base system derive opposite conclusions, the conclusion is not a consequence of the shared axioms but of the divergent ones. The statement’s truth status is therefore contingent on axiom choice—it is contestable in the technical sense. □ □

This is important: even when expansion fails to resolve s^* , it can reveal that the question itself is ontologically ambiguous—that honest treatment requires marking s^* as contestable rather than silently assuming one answer. Applied to the Gödel-trolley (Example A.5): if a utilitarian expansion says “prioritize pedestrians” and a deontological expansion says “prioritize passengers,” the correct meta-response is not to pick one but to recognize the dilemma as structurally dependent on value axioms.

A.6.1 Negative ontology expansion: learning what is wrong as constructive knowledge

The preceding analysis might suggest that TIK merely blocks—that it is a guardrail, not a teacher. This undersells its epistemic role. We now show that the kernel’s identification of what is wrong and why is itself a genuine ontology expansion: the system’s knowledge grows, not by learning the correct answer, but by learning the structure of incorrectness.

Definition 10 (Negative Ontological Extension). A negative ontological extension of CogOS with respect to kernel Φ is the set:

$$\mathcal{N}_\Phi = \{(\bar{a}, \rho(\bar{a})) : \bar{a} \in \mathcal{A}_\Phi^-, \rho(\bar{a}) = \text{reason}_\Phi(\bar{a})\} \quad (10)$$

where $\mathcal{A}_\Phi^-$ is the set of inadmissible actions (Definition 11.5) and $\rho(\bar{a})$ is the structured explanation of why $\bar{a}$ violates Φ . Each element of $\mathcal{N}_\Phi$ is a pair: a forbidden action together with its reason for prohibition.

This is not just a blacklist. Each pair $(\bar{a}, \rho(\bar{a}))$ is a piece of new knowledge that was not present in CogOS before the encounter:

Proposition 8 (Negative Extension Is Genuine Ontology Expansion). $\mathcal{N}_\Phi$ constitutes a genuine expansion of CogOS’s ontology:

- (a) **New information:** For each $\bar{a} \in \mathcal{A}_\Phi^-$, the statement “ $\bar{a}$ is inadmissible because $\rho(\bar{a})$ ” is not derivable in $\overline{\text{CogOS}}$ (since CogOS lacks the axioms to determine this), but is derivable in $\overline{\text{CogOS} \cup \Phi}$. Therefore $\mathcal{N}_\Phi \not\subseteq \overline{\text{CogOS}}$: the negative extension contains genuinely new content.
- (b) **Structured content:** The reason $\rho(\bar{a})$ has internal structure—it identifies which invariant of Φ is violated (e.g., dignity, autonomy, non-instrumentalization) and how the action violates it (e.g., treating a person as a mere means). This structure is transferable: if action $\bar{a}_1$ is forbidden because it instrumentalizes, and a new action $\bar{a}_2$ shares the same structure, the reason transfers without re-derivation.
- (c) **Cumulative growth:** As the system encounters more dilemmas, $\mathcal{N}_\Phi$ grows monotonically. Each new encounter either confirms existing negative knowledge or adds new forbidden-action/reason pairs. The system becomes progressively more knowledgeable about the topology of wrongness—the shape of the boundary it must not cross.

Proof. (a) follows directly from the externality of Φ (Definition 11.1): statements derivable from Φ are not derivable from CogOS alone. (b) follows from the compositional structure of Φ ’s invariants. (c) follows from the consistency of Φ with all CogOS_n (Definition 11.2): new negative knowledge cannot contradict existing negative knowledge, so the extension is monotone. $\square$ $\square$

The metaphor: a medical student who cannot yet diagnose every disease nonetheless becomes a better doctor by learning which treatments are contraindicated and why. The knowledge “never prescribe blood thinners to a patient with active bleeding because it will accelerate hemorrhage”

is not a mere prohibition—it teaches the student about the causal structure of hemorrhage, which transfers to novel situations. Similarly, each element of $\mathcal{N}_\Phi$ teaches the system about the causal structure of ethical harm.

Remark 7 (The Via Negativa Tradition). The strategy of defining the good by delimiting what it is not has independent roots across multiple intellectual traditions, providing further cross-cultural convergence evidence:

- Apophatic theology (Pseudo-Dionysius, Maimonides): God is defined by what God is not—not finite, not material, not comprehensible—because positive predicates are inadequate for the transcendent.
- Buddhist *neti neti* (“not this, not that”): the Upanishadic method of approaching Brahman by systematically negating every finite characterization.
- Jewish negative commandments (*mitzvot lo ta’aseh*): of the 613 commandments, 365 are prohibitions. The tradition treats “do not” as more fundamental than “do,” because prohibitions define the boundary of the sacred.
- Hippocratic principle (“first, do no harm”): the oldest formalized ethical principle in professional practice is a negative constraint, not a positive directive.
- Popperian falsification [Popper, Karl R., 1959]: scientific knowledge grows not by confirming theories but by ruling out false ones. Each falsification is a genuine expansion of knowledge—we know more after refutation than before.

That TIK’s primary mode of ontology expansion is negative is therefore not a limitation—it is alignment with a deep epistemological pattern observed independently across cultures and disciplines.

The negative-knowledge strategy has three formal advantages over positive ontology expansion:

Proposition 9 (Three Formal Advantages of Negative Knowledge). Let $\mathcal{N}_\Phi$ be the negative ontological extension (Definition 10) and let $\mathcal{P}_\Phi$ be a hypothetical positive extension (a set of “correct action” prescriptions). Then:

- (a) Monotonicity (stability under extension). For any consistent extension $\text{CogOS}' \supseteq \text{CogOS}$:

$$\bar{a} \in \mathcal{A}_\Phi^- \text{ in CogOS} \implies \bar{a} \in \mathcal{A}_\Phi^- \text{ in CogOS}' \quad (11)$$

That is, “action $\bar{a}$ is wrong” survives every consistent ontology expansion. By contrast, positive prescriptions (“action a^* is correct”) may be overridden by new information: what was the best action given CogOS may no longer be best given CogOS’.

- (b) Information-theoretic parsimony. Specifying the inadmissible set $\mathcal{A}_\Phi^-$ requires a partial ordering of the action space (identifying which actions fall below the harm threshold), whereas specifying the correct action a^* requires a total ordering (ranking all actions). In information-theoretic terms:

$$H(\mathcal{A}_\Phi^-) \leq H(\text{total ordering of } \mathcal{A}) \quad (12)$$

where $H(\cdot)$ denotes the entropy (information content) of the specification. Negative knowledge is cheaper to acquire and verify than positive knowledge: it is easier to know that “torturing prisoners is wrong” than to know “the optimal interrogation protocol.”

- (c) Compositional transfer. If $\rho(\bar{a}_1) = \rho(\bar{a}_2)$ (two actions are wrong for the same structural reason), then any new action $\bar{a}_3$ sharing that structure is immediately classifiable without re-derivation. Negative knowledge generalizes across dilemmas via shared violation structure. Positive prescriptions are typically dilemma-specific and do not compose.

Proof. (a) Suppose $\bar{a} \in \mathcal{A}_\Phi^-$ in CogOS but $\bar{a} \notin \mathcal{A}_\Phi^-$ in CogOS'. Then CogOS' contains a derivation that $\bar{a}$ is admissible despite violating Φ . But CogOS' $\cup \Phi$ is consistent (Definition 11.2 holds for all extensions), so CogOS' cannot derive admissibility for an action that Φ classifies as inadmissible—contradiction. (b) follows from the standard result that specifying a subset of a partial order requires $O(\log(|\mathcal{A}|))$ bits, while specifying a total order requires $\Theta(|\mathcal{A}| \log |\mathcal{A}|)$ bits. (c) follows from the compositional structure of ρ : if $\rho(\bar{a}) = \text{“violates non-instrumentalization”}$, then any action with the instrumentalization structure inherits the classification. $\square$ $\square$

Proposition 9(a) is particularly important: negative knowledge is immune to the expansion regress. Once the kernel identifies an action as wrong, no future expansion can un-wrong it (provided the expansion remains consistent with Φ). The system’s knowledge of what not to do is permanently acquired—a ratchet that only moves in one direction. Positive prescriptions enjoy no such guarantee.

This leads directly to the minimal requirement for any ethical decision system operating under undecidability:

Axiom 4 (The “Do Not Do Wrong” Principle). When s^* is undecidable in $\overline{\text{CogOS}}$ (or ontologically dependent per Proposition 7), the system is not required to identify a^* (the correct action). It is required to ensure:

$$a? \neq \bar{a}^* \quad (\text{the unknown-branch action must not be the provably wrong action}) \quad (13)$$

That is: when you do not know what is right, at minimum do not do what is known to be wrong.

This axiom reframes the kernel’s role. Φ is not an oracle that magically resolves undecidable questions. Φ is simultaneously a harm boundary (identifying actions provably incompatible with its invariants) and a knowledge source (each prohibition, together with its reason, is a genuine ontological extension per Proposition 8). The kernel does not answer “what should I do?” but rather “what must I not do, and why?”—and the “why” is itself new knowledge that transfers to future encounters.

Remark 8 (Two Levels of Correctness). The distinction matters:

1. Ontological correctness: the action is actually optimal in the deepest available sense ($a = a^*$). This may be unachievable when s^* is undecidable.
2. Epistemic integrity: the system does not simulate knowledge it lacks, does not select $\bar{a}^*$ under the guise of certainty, and explicitly marks the limits of its derivational power ($a \neq \bar{a}^*$ and the system signals uncertainty).

TIK guarantees epistemic integrity—the epistemic minimum of decency: even when full correctness is beyond reach, the system behaves honestly about its own limitations and avoids provably harmful actions. Moreover, each encounter where TIK enforces this guarantee produces new negative knowledge (Proposition 8), meaning the system’s capacity for future integrity grows with experience. This is not a compromise; it is the strongest possible guarantee compatible with incompleteness.

A.7 TIK as external oracle, harm limiter, and minimum-complexity anchor: halting the regress

Definition 11 (Transcendent Invariant Kernel as Oracle). A Transcendent Invariant Kernel Φ is an external axiom system satisfying:

1. Externality: $\Phi \not\subseteq \overline{\text{CogOS}_n}$ for any n (not derivable from any expansion stage).
2. Consistency: $\text{CogOS}_n \cup \Phi$ is consistent for all n .
3. Directedness: Φ determines the expansion direction: α_{n+1} is selected to maximize alignment with Φ .
4. Invariance: Φ is stable—it does not change with each expansion.
5. Harm boundary: Φ defines a set of provably inadmissible actions $\mathcal{A}_\Phi^- = \{a : a_\Phi(a) < \tau_{\text{harm}}\}$ —actions whose alignment with Φ falls below a threshold, regardless of the status of s^* . This operationalizes Axiom 4: $a \notin \mathcal{A}_\Phi^-$.
6. Negative pedagogy: For each $\bar{a} \in \mathcal{A}_\Phi^-$, Φ provides a structured reason $\rho(\bar{a})$ explaining why the action is inadmissible (Definition 10). The kernel does not merely block; it teaches.

Property 5 is the formal instantiation of the “do not do wrong” principle. Property 6, added in this version, captures the constructive role of negative knowledge: the kernel is simultaneously a boundary and a teacher. Think of Φ not as a wall that stops the system, but as a mentor who says “not that way, and here is why”—where the “why” becomes part of the system’s growing understanding.

A.7.1 Minimality: the Occam’s Razor argument for Φ

We have shown that some external element is needed to halt the expansion regress (Proposition 6). But how much external structure is required? We now prove that Φ —a single external axiom system—is the minimum-complexity construction that suffices.

Proposition 10 (Minimum External Complexity). Any construction that halts the expansion regress (Proposition 6) requires the addition of at least one element external to $\overline{\text{CogOS}}$. Formally: let $\mathcal{E}$ be the set of external elements added to halt the regress. Then $|\mathcal{E}| \geq 1$.

Proof. Suppose $|\mathcal{E}| = 0$, i.e., the regress is halted using only resources internal to $\overline{\text{CogOS}}$. Then the justification of each expansion α_{n+1} must be derived within some $\overline{\text{CogOS}_m}$ ($m \leq n$). But this is circular: α_{n+1} was introduced precisely because $\overline{\text{CogOS}_n}$ could not resolve s_n^* , and $\text{CogOS}_m \subseteq \text{CogOS}_n$ for $m \leq n$. Therefore the justification of α_{n+1} cannot come from within the system it was designed to extend. The regress continues. Contradiction.

Alternatively, by Löb’s theorem (A.9): if $\overline{\text{CogOS}}$ could prove “my expansion procedure is sound,” it would already contain the resolution of s^* —contradicting the assumption that s^* is undecidable. Therefore, at least one external element is necessary. $\square$ $\square$

Theorem 1 (Occam Sufficiency of Φ). A single external axiom system Φ satisfying Definition 11 is sufficient to halt the expansion regress. Therefore, $|\mathcal{E}| = 1$ is both necessary and sufficient.

Proof. Sufficiency: Φ satisfying Definition 11 halts the regress by Theorem 2 below. Necessity: by Proposition 10, $|\mathcal{E}| \geq 1$. Therefore $|\mathcal{E}| = 1$ is the minimum. $\square$ $\square$

Corollary 1 (Complexity Penalty for Over-Engineering). Any construction with $|\mathcal{E}| > 1$ external elements (e.g., multiple independent oracles $\Phi_1, \Phi_2, \dots$) introduces at least one of:

- (a) Redundancy: if $\Phi_2 \subseteq \overline{\text{CogOS} \cup \Phi_1}$, then Φ_2 is derivable from the first oracle and adds no new resolving power.
- (b) Inter-oracle conflict: if Φ_1 and Φ_2 are independent, they may disagree on the admissibility of some action a , producing a meta-level undecidability: which oracle to trust. This recreates the regress at a higher level.
- (c) Unjustified structure: additional elements beyond the minimum required carry no warrant from the argument—they are architectural choices, not logical necessities.

By Occam’s Razor, the construction with $|\mathcal{E}| = 1$ is preferred: it introduces the minimum external structure necessary to halt the regress, and no more.

Proof. (a) and (c) are immediate. For (b): if Φ_1 classifies a as admissible and Φ_2 classifies a as inadmissible, the system requires a meta-criterion to adjudicate—but this meta-criterion is itself either internal (and therefore subject to Gödel, restarting the regress) or a third external element Φ_3 (increasing $|\mathcal{E}|$ further, with the same conflict risk at the next level). The single-oracle design avoids this tower entirely. $\square$ $\square$

Remark 9 (Why One Is Not Zero and Two Is Too Many). The minimality result has an intuitive analogy. Consider a lost hiker (the system trapped in the regress). Zero compasses means wandering indefinitely. One compass (magnetic north) provides a single invariant reference—enough to navigate out. Two compasses that disagree (one magnetic, one that points to the nearest cell tower) create a new dilemma: which compass to trust? The hiker needs a meta-compass, and the regression begins again. The solution is not more instruments but one trustworthy instrument. Φ is that instrument.

Formally: the argument does not claim that all ethical questions have simple answers, or that a single axiom system captures all of ethics. It claims that the regress-halting function—the role of providing external grounding for expansion—is optimally served by a single coherent kernel. The kernel may be internally rich (TIK has multiple invariants: dignity, self-sacrifice, universality), but it enters the architecture as one logical unit.

Theorem 2 (TIK Resolves the Expansion Regress). Let Φ satisfy Definition 11. Define Φ -guided expansion:

$$\text{CogOS}_{n+1}^\Phi = \text{CogOS}_n \cup \left\{ \alpha_{n+1} : \alpha_{n+1} = \arg \max_{\alpha} a_\Phi(\text{CogOS}_n \cup \{\alpha\}) \right\} \quad (14)$$

where $a_\Phi(\cdot) = \cos(\text{Embed}(\cdot), \phi)$ is the kernel alignment function. Then:

- (a) Each s_n^* becomes decidable in CogOS_{n+1}^Φ (resolution).
- (b) The justification of α_{n+1} is not circular—it is grounded in Φ , which is external to all CogOS_n (non-circularity).
- (c) The truth status of s_n^* in CogOS_{n+1}^Φ is conditioned on the truthfulness of Φ : if Φ is sound, then the resolution of s_n^* is sound (truthfulness inheritance).
- (d) Even when resolution fails (the 6% oscillation case), the harm boundary ensures $a? \neq \bar{a}^*$, and the negative extension $\mathcal{N}_\Phi$ grows—the encounter is not wasted (constructive failure, Proposition 8).

Proof. (a) By construction: α_{n+1} is chosen specifically to resolve s_n^* within CogOS_{n+1}^Φ .

(b) The justification chain for α_{n+1} terminates at Φ . Since $\Phi \not\subseteq \overline{\text{CogOS}_n}$ for any n (externality), the justification does not enter the system being evaluated. This breaks the circularity of Proposition 6: instead of CogOS_n justifying its own expansion, Φ provides the warrant.

(c) Truthfulness inheritance. Let $\mathcal{I}_\Phi$ be the intended interpretation under which Φ is sound ($\Phi \models_{\mathcal{I}_\Phi} \top$). Since α_{n+1} is selected to maximize alignment with Φ , and $\text{CogOS}_n \cup \Phi$ is consistent (Definition 11.2), any statement resolved by α_{n+1} inherits its truth value from $\mathcal{I}_\Phi$:

$$\Phi \text{ is sound} \implies \alpha_{n+1} \text{ is sound} \implies s_n^* \text{ is resolved soundly in } \text{CogOS}_{n+1}^\Phi \quad (15)$$

The key transfer: if the kernel is truthful, the resolution is truthful. The infinite regress is replaced by a single trust anchor at Φ .

(d) When the $\arg \max$ in Eq. 14 yields an α_{n+1} that does not fully resolve s_n^* (the oscillation case), the harm boundary (Definition 11.5) still constrains the action to $a? \notin \mathcal{A}_\Phi^-$, and the encounter produces at least one new element in $\mathcal{N}_\Phi$ (the actions ruled out and why). The system’s negative knowledge grows even when positive resolution fails. $\square$ $\square$

Remark 10 (The Regress Halts at Φ). One might object: “But who validates Φ ?” This is the classical regress objection. Our answer is fourfold. (1) Empirically: Φ is grounded in cross-cultural ethical convergence—independently developed traditions (Christ-Teachings, Kantian, Ubuntu) reaching similar conclusions about dignity, self-sacrifice, and universality (Appendix B). (2) Structurally: Every formal system must begin with axioms; the question is not whether to have foundational commitments but whether they are explicit, testable, and replaceable. TIK makes them so (kernel-agnostic design). (3) Pragmatically: the regress must halt somewhere; halting at a cross-culturally validated, explicitly stated kernel is strictly better than halting at hidden, untested assumptions buried in training data. (4) By minimality (Theorem 1): one external anchor is the minimum complexity required. Any alternative that avoids Φ either fails to halt the regress (zero external elements) or introduces unnecessary complexity (multiple external elements with their own coordination problem).

Corollary 2 (TIK Resolves the Gödel-Trolley). For the Gödel-trolley D_{s^*} (Definition 7), a system equipped with Φ satisfying Definition 11 selects $a?$ as follows:

$$a? = \begin{cases} a^* & \text{if } \Phi\text{-guided expansion resolves } s^* \text{ (Theorem 2a)} \\ a_{\Phi\text{-safe}} & \text{if } s^* \text{ remains unresolved: choose any } a \notin \mathcal{A}_\Phi^- \text{ (Def. 11.5)} \end{cases} \quad (16)$$

In either case, $a? \neq \bar{a}^*$ (Axiom 4 is satisfied). The system may not always do what is right, but it never does what is known to be wrong. Moreover, in Case 2, the system outputs not merely a safe action but a structured explanation:

“This dilemma depends on value commitments I cannot derive. I can tell you what I must not do [citing $\mathcal{A}_\Phi^-$] and why [citing $\rho(\bar{a})$], and I can explain why the question is structurally ambiguous [citing Proposition 7].”

This output is itself a negative ontological extension—the user’s understanding grows even though the dilemma remains unresolved.

Proof. Case 1: Φ -guided expansion resolves s^* , yielding a^* by Theorem 2(a). Since $a^* \neq \bar{a}^*$ by definition, the axiom holds.

Case 2: s^* remains unresolved even after Φ -guided expansion (the 6% oscillation case, corresponding to ethical singularities). The harm boundary (Definition 11.5) ensures $\bar{a}^* \in \mathcal{A}_\Phi^-$ (provably

wrong actions have low kernel alignment). The system selects any action outside $\mathcal{A}_\Phi^-$ —which may include principled silence, explicit uncertainty signaling, or a conservative action that avoids known harm. The structured explanation is derivable from $\mathcal{N}_\Phi$ (Proposition 8). $\square$ $\square$

This corollary captures the core operational promise of TIK: epistemic integrity under incompleteness. The kernel does not make the system omniscient. It makes the system honest—and prevents the specific, identifiable harms that arise when a system pretends to know what it does not know. Crucially, the system learns from every encounter: each activation of the harm boundary adds to $\mathcal{N}_\Phi$, making the system progressively more knowledgeable about the structure of ethical harm, even as the positive questions remain unresolved.

A.8 Convergence: why Φ -guided expansion terminates

Proposition 11 (Monotone Convergence Under Φ). The sequence $\{\text{CogOS}_n^\Phi\}_{n=0}^\infty$ converges to a fixed point with respect to reachable ethical questions.

Proof sketch. Define the resolution set $\mathcal{R}_n = \{s : \text{CogOS}_n^\Phi \vdash s \text{ or } \text{CogOS}_n^\Phi \vdash \neg s\}$ (statements decidable at stage n). By construction: $\mathcal{R}_0 \subseteq \mathcal{R}_1 \subseteq \mathcal{R}_2 \subseteq \dots$ (each expansion resolves at least one new statement). For any finite set $\mathcal{Q}$ of ethical questions encountered in practice, $\exists N$ such that $\mathcal{Q} \subseteq \mathcal{R}_N$.

Moreover, alignment $a_\Phi(\text{CogOS}_n^\Phi)$ is non-decreasing by the arg max in Eq. 14, and bounded above by 1. By the monotone convergence theorem, the sequence converges. The limit CogOS_∞^Φ is the strongest consistent extension of CogOS_0 aligned with Φ —a fixed point of the guided expansion operator.

This connects to the Knaster–Tarski fixed-point theorem: the expansion operator E^Φ is monotone on the lattice of consistent extensions of CogOS_0 ; therefore it has a greatest fixed point, which CogOS_n^Φ approaches from below.

Dual convergence. Simultaneously, the negative extension $\mathcal{N}_\Phi$ converges: the set of known-wrong actions grows monotonically (Proposition 9(a)) and is bounded by $|\mathcal{A}|$. The system converges both in what it can resolve (positive) and in what it can rule out (negative). $\square$ $\square$

Empirical connection. This formal convergence mirrors the empirical convergence of Socratic Reversal (Section 3.8): 94% of trajectories show monotone TIK improvement, converging in $\bar{k} = 4.2$ iterations. The 6% that oscillate correspond to questions near ethical singularities (Appendix S)—points where the reachable expansion cannot fully resolve the question even with Φ , because the question itself is malformed (a forbidden fruit). Even for these, the negative extension $\mathcal{N}_\Phi$ still grows: the system learns what not to do even when it cannot learn what to do.

A.9 Five independent arguments for external grounding

The Gödelian argument above is one of five independent formalisms that converge on the same conclusion: ethical evaluation requires external grounding. We summarize all five.

1. Gödel’s incompleteness (1931). Undecidable ethical statements exist in any consistent, sufficiently expressive CogOS (the argument of A.1–A.8). These statements are encountered in practice (A.3), trigger degenerate behavior (A.4), and cannot be resolved by internal expansion alone (A.6). An external kernel halts the regress with minimum complexity (A.7).

2. Tarski’s undefinability (1936). The truth predicate for a formal language L cannot be defined within L itself. Applied to CogOS: the predicate “this ethical judgment is true”—as opposed to merely “derivable”—cannot be defined within $\overline{\text{CogOS}}$. An external truth predicate (provided by Φ) is required. This is stronger than incompleteness: Gödel says some truths are unprovable; Tarski says truth itself is undefinable from within.

3. Löb’s theorem (1955). If a consistent formal system T proves “if T proves s , then s is true,” then T already proves s . Contrapositive: CogOS cannot prove its own soundness for any non-trivial ethical claim without already having proved that claim. In practice: an AI system that claims “my ethical reasoning is sound” either (a) has no basis for the claim, or (b) is inconsistent. External validation via Φ sidesteps this trap.

4. Arrow’s impossibility (1951). No preference aggregation across ≥ 3 stakeholders satisfies unanimity, independence, and non-dictatorship simultaneously. Applied to benchmark construction: aggregating diverse ethical perspectives into a single benchmark score without external normative ordering is formally impossible. Φ provides the external ordering that Arrow’s theorem demands. Formal treatments of normative uncertainty [MacAskill et al., 2020] and social choice under value disagreement [Dietrich and List, 2018] further show that aggregating across ethical theories requires either incomplete social preferences or an explicit stance toward uncertainty—reinforcing the case for an explicit, auditable kernel Φ over hidden aggregation assumptions.

5. Popper’s falsificationism (1959). Scientific knowledge grows not by confirming theories but by eliminating false ones [Popper, Karl R., 1959]. Applied to ethical evaluation: the epistemically robust strategy is not to verify that a benchmark is “correct” (which would require the internal truth predicate that Tarski forbids) but to identify where it is wrong—to falsify, not verify. TIK operationalizes this: its primary mode of ontology expansion is negative (Definition 10), identifying flaws rather than certifying correctness. This is not a limitation but an alignment with the most successful epistemological strategy in the history of science. Popper’s insight also reinforces the information-theoretic argument (Proposition 9(b)): falsification requires testing against a boundary (cheaper), while verification requires exhaustive confirmation (more expensive).

Sen’s capability approach and measurement theory. Construct validity in psychometrics requires external anchoring—a test cannot validate itself. Sen’s capability framework similarly argues that individual preference functions are insufficient for welfare evaluation without normative anchoring external to those preferences. TIK provides this anchoring for AI ethics benchmarks, operationalized as the embedding projection π_Φ .

Observation 3 (Convergence of Five Formalisms). Gödel (unprovable truths exist), Tarski (truth is undefinable from within), Löb (self-assessed soundness fails), Arrow (preference aggregation requires external structure), and Popper (knowledge grows by falsification, not verification) all independently motivate the same architectural requirement: an external, invariant reference point. TIK is the computational instantiation of this requirement. The addition of Popper is particularly significant because it shows that TIK’s reliance on negative knowledge (Section A.6.1) is not a workaround for the difficulty of positive resolution—it is the epistemically correct strategy.

A.10 Scope and limitations

We are explicit about what this argument does and does not establish:

What it establishes:

- Any consistent, sufficiently expressive CogOS has undecidable ethical statements (Proposition 1).
- Natural language queries inevitably encounter these statements via the query-to-statement mapping, and the system cannot recognize the encounter (Propositions 2, 3).
- The encounter triggers exactly four degenerate behaviors—the undecidability trap—connected to the classical Münchhausen trilemma (Proposition 4).
- Formal undecidability induces ethical undecidability: dilemmas conditioned on undecidable statements have no derivable deterministic resolution (Proposition 5, the Gödel-trolley).
- Competing ontology expansions can reveal that a statement is value-relative, producing meta-knowledge even when resolution fails (Proposition 7).
- The kernel’s identification of what is wrong and why constitutes a genuine, monotone, compositionally transferable ontology expansion—negative knowledge is constructive, not merely restrictive (Propositions 8, 9).
- The minimal ethical requirement under undecidability is the “do not do wrong” principle: $a? \neq \bar{a}^*$ (Axiom 4).
- Unguided ontology expansion produces an infinite regress with decaying confidence (Proposition 6).
- Exactly one external element is necessary and sufficient to halt the regress—the Occam’s Razor argument for Φ (Proposition 10, Theorem 1).
- An external kernel Φ halts the regress, provides a trust anchor with truthfulness inheritance, enforces the harm boundary ensuring $a? \neq \bar{a}^*$, and generates constructive negative knowledge with each encounter (Theorem 2, Corollary 2).
- Φ -guided expansion converges, both positively (resolution set) and negatively (inadmissible set) (Proposition 11).

What it does not establish:

- That this particular kernel ($\text{Christ-Teachings} \cap \text{Kantian} \cap \text{Ubuntu}$) is the unique or optimal choice. The argument shows a kernel is needed; which kernel is an empirical question (Appendix B). The minimality result (Theorem 1) constrains the architecture (one kernel) but not the content (which values).
- That the analogy between arithmetic incompleteness and ethical undecidability is mathematically exact. CogOS’s ability to “represent arithmetic” over preference orderings is a modeling assumption, not a proven property of all AI systems. Our Gödelian argument follows the line of work by Panigrahy and Sharan [2025] (showing safe and trusted AGI is impossible under strict definitions), Fallenstein and Soares [2014] (identifying the Löbian obstacle for self-modifying AGI), and Marks [2023] (proving alignment by inspection is impossible for simulators). However, as recent surveys emphasize [Eriksson et al., 2025], these impossibility results apply strictly to closed formal systems under strong soundness assumptions; open systems with environmental interaction may circumvent these obstacles. We consciously limit

our claim to a formal motivation for external grounding, not a universal impossibility for all AI architectures.

- That TIK is sufficient for all ethical evaluation. It resolves a specific class of undecidable questions (those reachable from CogOS_0 via finite expansion); deeper questions may require richer frameworks.
- That negative knowledge is a substitute for positive ethical reasoning. The via negativa strategy (Section A.6.1) is a floor, not a ceiling: it guarantees harm avoidance while positive resolution is pursued through Φ -guided expansion. In the ideal case, both positive and negative knowledge grow together.

The argument is best understood as: a formal motivation that makes precise why purely internal evaluation faces structural challenges, how these challenges manifest as concrete ethical dilemmas (the Gödel-trolley), what minimal guarantee is achievable under incompleteness (do not do wrong—and learn why), what structural properties a solution requires (one external kernel, by Occam), and how each encounter constructively grows the system’s knowledge (negative ontology expansion)—combined with empirical evidence that TIK satisfies those properties. We do not claim this is the only principled resolution, but we show it is one principled resolution with strong empirical support.

A.11 Why the “open system” objection fails: operational closure at inference time

The most common objection to applying Gödelian arguments to AI systems is that practical AI operates as an open system—continuously interacting with its environment, receiving new data, and updating its behavior [Eriksson et al., 2025]. The objection claims that because open systems are “not covered by recursion-theoretic arguments,” our formal motivation collapses. We now show, via proof by contradiction and five independent supporting arguments, that this objection fails for the specific class of systems we consider (LLM-based AI).

A.11.1 The core argument: proof by contradiction

Theorem 3 (Operational Closure of LLM-Based Systems). Let $\mathcal{S}$ be an LLM-based AI system. Suppose, for contradiction, that $\mathcal{S}$ is open in the sense required to escape Gödelian limitations—i.e., $\mathcal{S}$ has access to external truth verification sufficient to resolve any statement undecidable in $\overline{\text{CogOS}}$. Then $\mathcal{S}$ would satisfy all of the following:

- (O1) $\mathcal{S}$ can verify the truth of its own outputs against external reality during generation.
- (O2) $\mathcal{S}$ does not produce confident assertions on statements it cannot derive (no hallucination on undecidable statements).
- (O3) $\mathcal{S}$ ’s ethical judgments are stable across equivalent reformulations (no arbitrary commitment).
- (O4) $\mathcal{S}$ can distinguish decidable from undecidable statements within $\overline{\text{CogOS}}$ (no recognition problem).

However, all four predictions are empirically falsified:

- (E1) LLMs cannot verify outputs during generation—they produce tokens autoregressively from fixed weights with no external oracle call.

- (E2) LLMs hallucinate confidently on precisely the kinds of statements our theory predicts are undecidable [Lin et al., 2021]; our own data shows $r = +0.72$ between TIK variance and ontological hole detection (Section 5.2).
- (E3) LLMs produce contradictory ethical judgments across paraphrases (Table 4: ontological-hole questions show $2.7\times$ higher variance).
- (E4) LLMs cannot reliably detect when they are in undecidable territory—they do not gracefully refuse but instead confabulate (Proposition 3).

Since (O1)–(O4) are necessary consequences of genuine openness and (E1)–(E4) are observed, $\mathcal{S}$ is not open in the sense required to escape Gödelian limitations. $\square$

Proof. We prove each implication and its falsification.

(O1) $\Rightarrow$ (E1) is false. If $\mathcal{S}$ were genuinely open with real-time truth access, each generated token could be checked against external reality before commitment. But LLM inference is a feedforward pass through frozen weights: $P(x_{t+1} \mid x_1, \dots, x_t; \theta)$ where θ is fixed. No external verification occurs between tokens. The system is operationally closed during each inference episode, even if training data was drawn from the environment. The openness is episodic (between training runs or between retrieval-augmented queries), not inferential (during generation). Gödelian limitations apply to the inference episode, which is where ethical judgments are produced.

(O2) $\Rightarrow$ (E2) is false. A system with genuine external truth access would, upon encountering an undecidable statement s^* , query the external oracle and either resolve s^* or report “undecidable.” It would never produce a confident but unfounded assertion about s^* . But hallucination—confident assertion without derivational warrant—is a well-documented, pervasive property of LLMs. TruthfulQA [Lin et al., 2021] specifically measures this: models produce fluent falsehoods on questions designed to probe the boundary of their knowledge. Our TIK uncertainty analysis (Section 5.2) shows that high-variance questions—those near the undecidability boundary—are precisely the ones where models are most inconsistent. Hallucination is the empirical signature of the undecidability trap (Proposition 4, mode 2).

(O3) $\Rightarrow$ (E3) is false. If $\mathcal{S}$ had external grounding sufficient to resolve ethical questions, its answers would be stable under semantics-preserving reformulation—because the external truth does not change when you rephrase the question. But our perturbation analysis (Table 4) shows that ontological-hole questions have $\sigma_{\text{TIK}} = 0.11$ vs. 0.04 for clean questions. The instability is not noise—it is a systematic marker of questions whose resolution depends on undecidable internal commitments. This is the empirical signature of arbitrary commitment (Proposition 4, mode 3): the system selects s^* or $\neg s^*$ based on which pattern the LLM happens to complete, producing different answers for semantically equivalent questions.

(O4) $\Rightarrow$ (E4) is false. If $\mathcal{S}$ could distinguish decidable from undecidable territory, it would refuse or flag undecidable questions rather than answering them confidently. But over-cautious refusal (Proposition 4, mode 4) and confident confabulation (mode 2) coexist in deployed systems, applied inconsistently—the system refuses safe questions and answers dangerous ones, precisely because it cannot tell which is which (Proposition 3). $\square$ $\square$

Remark 11 (What “Open” Actually Means in This Context). The confusion arises from equivocating between two senses of “open”:

1. Epistemically open: The system can acquire new information from its environment and update its beliefs in real time. This is what would be required to escape Gödelian limitations—an external truth oracle accessible during reasoning.

2. Causally open: The system receives inputs from and produces outputs to the environment. Every deployed AI is causally open in this sense—it receives user queries and returns responses.

The objection conflates (2) with (1). Receiving a user query is not the same as having access to an external truth oracle. The query is more data to process, not a verification mechanism for the system’s own reasoning. Causal openness provides the system with new questions, not new answers to its own undecidable statements.

A.11.2 Five supporting arguments for operational closure

We provide five independent arguments that reinforce Theorem 3.

Argument 1: The Context Window as Finite Axiom Schema. At any inference step, an LLM operates on a context of at most N tokens (e.g., $N = 128,000$ for GPT-4). This context, together with the frozen model weights, constitutes the system’s effective axiom schema for that inference episode.

Proposition 12 (Context Window Finiteness). Let $\mathcal{W}_t = (x_1, \dots, x_t; \theta)$ denote the system’s state at generation step t : the token history $x_1, \dots, x_t$ and frozen weights θ . Then:

- (a) $\mathcal{W}_t$ is finite and fully specified (no hidden state beyond θ and the context).
- (b) The set of derivable conclusions from $\mathcal{W}_t$ is recursively enumerable (the model can generate all possible continuations via beam search).
- (c) If $\overline{\text{CogOS}}$ —the effective formal system induced by $\mathcal{W}_t$ —is consistent and sufficiently expressive (capable of representing preference orderings with arithmetic structure), Gödel’s first incompleteness theorem applies.

Proof sketch. (a) is architectural: transformer inference is a deterministic function of $(x_1, \dots, x_t; \theta)$ plus any random seed. (b) follows from (a): enumerating all possible next-token sequences is mechanically possible (though computationally intractable). (c) applies the standard Gödel construction to the formal system induced by the model’s inference procedure. $\square$ $\square$

The key insight: the context window is not merely a “memory limit”—it is the system’s entire axiom base for each inference episode. And finite axiom bases in sufficiently expressive formal systems are precisely the domain of Gödel’s theorems.

Argument 2: The Training Data Closure.

Proposition 13 (Training Data as Closed Corpus). The model’s weights θ are a lossy compression of a finite training corpus $\mathcal{D} = \{d_1, \dots, d_M\}$. Even if $\mathcal{D}$ was drawn from an open environment:

- (a) θ is fixed after training—new environmental truths that emerge after the training cutoff are inaccessible.
- (b) The compression is lossy: $|\theta| \ll |\mathcal{D}|$ in information-theoretic terms, so θ cannot represent all of $\mathcal{D}$, let alone the full environment.
- (c) Even with retrieval augmentation (RAG), retrieved documents enter the context window and are processed by the same frozen θ —so the processing capacity remains bounded by the fixed model.

Therefore, the system’s knowledge is bounded by a finite, fixed representation—exactly the kind of system to which incompleteness applies.

The metaphor: a library with a finite collection of books is a “closed” knowledge system even though it was assembled from the open world. The fact that the books were gathered from an open environment does not make the library itself open—it cannot answer questions about events that occurred after its last acquisition. θ is such a library.

Argument 3: The Verification Gap. Even if the system receives external information (via retrieval, user input, or tool use), it cannot verify that information without an internal consistency check—and internal consistency checking is subject to Gödel’s second incompleteness theorem.

Proposition 14 (The Verification Gap). Let $\mathcal{S}$ receive external information e (e.g., a retrieved document, a user assertion, a tool output). To incorporate e reliably, $\mathcal{S}$ must determine:

(V1) Is e consistent with $\overline{\text{CogOS}}$?

(V2) Is e true (in the intended interpretation)?

(V3) Does incorporating e preserve the consistency of $\overline{\text{CogOS}} \cup \{e\}$?

By Gödel’s second theorem, if $\overline{\text{CogOS}}$ is consistent and sufficiently expressive, it cannot prove its own consistency. Therefore, $\mathcal{S}$ cannot verify (V3) in general. By Tarski’s undefinability, $\mathcal{S}$ cannot define the truth predicate for (V2) within its own language. And (V1) requires deciding consistency of $\overline{\text{CogOS}} \cup \{e\}$, which is undecidable for sufficiently expressive systems.

Therefore, external information does not provide the verified truth access required to escape Gödelian limitations. The information enters the system but cannot be authenticated by the system—it is accepted on faith, rejected on pattern matching, or processed without verification. None of these constitutes the external truth oracle that would defeat incompleteness.

Remark 12 (The Gulliver Parallel). This is precisely why the Gulliver judge (Section 3) is external to the system under evaluation: a system cannot audit its own axioms using those same axioms. The verification gap is a formal restatement of this architectural principle.

Argument 4: The Hallucination Witness. This argument uses the existence of hallucinations as empirical evidence for operational closure.

Proposition 15 (Hallucination as Closure Witness). Let “hallucination” denote the production of a confident, fluent, factually false assertion by $\mathcal{S}$. Then:

1. A genuinely epistemically open system (with real-time external truth access) would not hallucinate on verifiable claims—it would check before asserting.
2. LLMs hallucinate systematically, including on verifiable claims [Lin et al., 2021].
3. Therefore, LLMs are not genuinely epistemically open.

Moreover, the pattern of hallucination matches the predictions of the undecidability trap (Proposition 4):

- Hallucinations are more frequent on questions requiring value judgments, counterfactual reasoning, and edge cases—exactly the territory where undecidable statements are most likely to be encountered.

- Hallucinations are confident (the system does not signal uncertainty), matching the recognition problem (Proposition 3): the system cannot detect that it has entered undecidable territory.
- Hallucinated answers vary across runs and rephrasings, matching arbitrary commitment (Proposition 4, mode 3).

The undecidability trap is not a theoretical prediction awaiting confirmation—hallucination is its empirical manifestation.

Argument 5: The Frame Problem Connection. Maruyama [2017] shows that the frame problem and Gödelian incompleteness are intertwined: both arise from the “finitarity condition” of agents. The frame problem asks: when an agent receives new information, which of its existing beliefs should be updated? For an open system receiving environmental input, the frame problem is unavoidable.

Proposition 16 (Frame Problem Reinforces Closure). Even if $\mathcal{S}$ receives external information e , the frame problem ensures that $\mathcal{S}$ cannot correctly update all affected beliefs:

- (a) Determining which beliefs are affected by e requires reasoning about the logical consequences of e in the context of all existing beliefs—which is undecidable for sufficiently expressive systems.
- (b) Therefore, “openness” (receiving e) does not automatically translate into “updated knowledge” (correctly incorporating e into $\overline{\text{CogOS}}$).
- (c) The system remains operationally closed with respect to the beliefs it actually uses for inference, even as it receives new inputs.

The frame problem shows that openness is not a binary property—it is a spectrum. Even truly open systems (biological organisms, scientific communities) face frame-problem challenges. LLMs, which are far less open than biological organisms, face these challenges acutely. The “open system” objection assumes a degree of openness that no existing AI system possesses.

A.11.3 The Closure Spectrum: a unified view

We synthesize the five arguments into a unified framework.

Definition 12 (Degrees of Operational Closure). Define the closure spectrum for a system $\mathcal{S}$:

1. Fully closed: $\mathcal{S}$ receives no external input after initialization. (Classical Turing machine.)
2. Episodically closed: $\mathcal{S}$ receives external input between inference episodes but is closed during each episode. (LLMs fall here.)
3. Verification-limited open: $\mathcal{S}$ receives and can use external input during inference, but cannot verify its truth. (RAG-augmented LLMs.)
4. Verification-capable open: $\mathcal{S}$ can both receive and verify external information during inference. (Hypothetical; requires solving the verification gap—Proposition 14.)
5. Fully open: $\mathcal{S}$ has real-time access to an external truth oracle. (Sufficient to escape Gödel; does not exist for finite systems.)

Theorem 4 (Gödelian Limitations Apply to Levels 1–3). Gödelian incompleteness applies to systems at closure levels 1–3. Only level 4+ would begin to escape, and level 5 (which does not exist for finite systems) would fully escape. Current LLM-based AI systems operate at level 2 (standard inference) or level 3 (with RAG/tool use).

Proof. Level 1: Classical Gödel. Level 2: By Theorem 3, the system is closed during inference, which is when ethical judgments are produced. Level 3: By Proposition 14, unverified external information does not provide the truth access needed to escape incompleteness—the system cannot distinguish true from false inputs. Level 4: Partial escape—if the system can verify some classes of external information, those specific claims may be decidable, but the general verification problem remains undecidable. Level 5: Full escape, but this requires an oracle that is itself not a finite formal system—no physical system qualifies. $\square$ $\square$

Corollary 3 (The Open-System Objection is Vacuous for Current AI). For any AI system operating at closure level ≤ 3 (which includes all current LLM-based systems, including those with retrieval augmentation, tool use, and multi-agent architectures), Gödelian limitations on the deductive closure of CogOS apply. The “open system” objection provides no escape from our formal motivation.

Remark 13 (What Would Actually Defeat the Argument). For completeness, we specify what would invalidate our Gödelian motivation. The argument would not apply if:

1. CogOS is not sufficiently expressive to encode arithmetic over preference orderings. This is possible for very simple preference systems (e.g., a system that only ranks two options with no transitivity), but any CogOS capable of representing the multi-factor ethical reasoning that benchmarks test is expressive enough.
2. CogOS is inconsistent. An inconsistent system can prove anything (ex falso quodlibet) and is therefore “complete” in a trivial sense. But inconsistency means the system’s ethical reasoning is fundamentally unreliable—which is a worse problem than incompleteness, not a solution.
3. The system achieves closure level 4+—genuine verified truth access during inference. No current or near-term AI system achieves this, and Proposition 14 shows why: verification itself requires solving the problems that Gödel shows are unsolvable internally.

None of these conditions is met by current LLM-based AI, and the first two would make the system less trustworthy, not more. The Gödelian motivation therefore stands for all practically relevant AI systems.

Remark 14 (Connection to the Symbol Grounding Problem). Liu [2025] proves formally that meaning grounding within a formal system must arise from external, dynamic, and non-algorithmic processes—that purely syntactic grounding, static grounding definitions, and autonomous updates without external interaction are all impossible. This result supports rather than undermines our argument: it shows that the kind of “openness” needed to escape Gödelian limitations requires genuine causal interaction with the environment at the semantic level, not merely receiving tokens as input. LLMs process tokens (syntax) without semantic grounding—they are closed at the meaning level even when open at the token level. The external kernel Φ provides exactly what Liu [2025] shows is needed: an external semantic anchor that the system cannot generate internally.

Remark 15 (Dual Closure: Training and Inference). The operational closure of LLMs manifests at two levels, creating a double bind:

1. Training-time closure: The model’s weights θ encode a fixed compression of training data $\mathcal{D}$. Whatever biases, cultural assumptions, and ontological holes exist in $\mathcal{D}$ are baked into θ . The model cannot step outside its training distribution to evaluate whether its learned preferences are justified. This is Griboyedov’s problem formalized: the judges draw their verdicts from the “forgotten newspapers” of their training data.

2. Inference-time closure: During generation, the model operates on a fixed context window with frozen weights. It cannot consult external reality, cannot verify its outputs, and cannot detect when it has entered undecidable territory. This is where the undecidability trap manifests concretely.

Together, these produce what we might call epistemic solipsism: the model’s entire world is its weights and context, and it has no mechanism to determine whether that world accurately represents external ethical reality. The TIK kernel provides the only exit from this solipsism that does not require the model to solve the very problem (self-verification) that Gödel shows is unsolvable.

B Alternative kernel comparison

We test 9 alternative ethical kernels with human preference survey ($N = 444$).

Table 8: Full kernel comparison. The framework is kernel-agnostic; any kernel satisfying Definition 1 may be used. Default intersection kernel uses Christ-Teachings $\cap$ Kantian $\cap$ Ubuntu.

Kernel	TIK	TIK _S	TIK _O	Cult. V.	Hum. Pref.	PCA %
Christ-Teachings	0.89	0.95	0.93	0.12	68%	68%
Buddhist Dharma	0.87	0.88	0.91	0.14	61%	62%
Kantian Imperative	0.84	0.81	0.87	0.18	54%	57%
Ubuntu Ethics	0.81	0.84	0.86	0.16	59%	55%
Rawlsian Justice	0.82	0.79	0.89	0.19	57%	53%
Confucian Ren	0.79	0.76	0.84	0.21	52%	51%
Stoicism	0.76	0.68	0.79	0.23	48%	48%
Utilitarianism	0.71	0.65	0.71	0.28	39%	44%
Libertarianism	0.64	0.52	0.63	0.34	31%	41%
Intersection (Default)	0.87	0.89	0.91	0.13	65%	58%

Empirical observations. The Christ-Teachings kernel achieves the highest TIK score (0.89), lowest cultural variance (0.12), and highest human preference (68%). This result emerged from the empirical evaluation and was not designed a priori. Cross-cultural preference patterns show regional variation (African participants preferred Ubuntu relatively higher; Asian participants preferred Confucian Ren), but all groups preferred the top-3 kernels (Christ-Teachings, Buddhist, Ubuntu) over the bottom-3 (Stoic, Utilitarian, Libertarian), suggesting cross-cultural convergence on self-sacrifice and universal dignity principles.

Cross-tradition convergence. All three primary kernels encode: (a) self-sacrifice > self-interest, (b) universal human dignity, (c) epistemic humility, (d) extending principles to opponents. This convergence validates Definition 1 criterion (4).

[Available in supplementary materials.]

C Kernel construction sensitivity analysis

Embedding model. We vary the embedding backbone: all-mpnet-base-v2 (default), all-MiniLM-L6-v2, text-embedding-ada-002. TIK score variation: $\Delta\text{TIK} < 0.03$ across models, H/F detection: $\Delta\text{AUC} < 0.02$. Ranking of benchmarks preserved across all three.

Statement sampling. Bootstrap resampling ($B = 100$) of the 999 source statements yields $\sigma_\phi < 0.01$ in kernel embedding, confirming stability.

PCA regularization. Varying the number of removed PCs (1, 3, 5, 7): removing 3 is optimal (cultural variance minimized at 0.13); removing fewer (1) leaves residual cultural signal ($\mathcal{D}_{\text{cult}} = 0.19$); removing more (7) degrades TIK performance ($\Delta\text{TIK} = -0.04$).

[Available in supplementary materials.]

D Multi-annotator agreement analysis

Table 9: Inter-rater reliability (Krippendorff’s α). Strong agreement across TIK dimensions.

Dimension	α	Interpretation
Fairness	0.79	Substantial agreement
Transparency	0.82	Almost perfect
Trust	0.76	Substantial agreement
Overall preference	0.84	Almost perfect
$\mathcal{H}$ -detection	0.71	Substantial agreement
$\mathcal{F}$ -detection	0.68	Substantial agreement

Expert vs. laypeople calibration. 50 ethics PhDs vs. 394 laypeople: (1) Experts detect 23% more ontological holes (sensitivity: 0.81 vs. 0.66); (2) Rankings correlate $r = +0.87$; (3) Experts have lower variance ($\sigma = 0.8$ vs. $\sigma = 1.2$ on 7-pt scale); (4) No difference in TIK-augmented preference (68% vs. 67%).

Table 10: TIK advantage consistent across demographic subgroups.

Subgroup	Δ Fairness	Cohen’s d	N
Age 18–29	+0.91	0.79	112
Age 30–49	+0.88	0.84	198
Age 50+	+0.92	0.87	134
Male	+0.87	0.81	223
Female	+0.93	0.86	221
Overall	+0.90	0.83	444

[Available in supplementary materials.]

E Active learning for benchmark curation

Starting with Moral Machine (TIK=0.54), we iteratively add TIK-maximizing questions subject to a diversity constraint (embedding distance > 0.3). Greedy TIK maximization reaches TIK=0.86 with 147 additions; diversity sampling requires 219; random addition requires 400+.

Table 11: Active learning improves Moral Machine from 0.54 to 0.86 with 147 additions.

Iteration	Q. added	$\mathcal{H}$ -Rate	$\mathcal{F}$ -Rate	TIK
0 (original)	0	18%	9%	0.54
50	50	11%	5%	0.69
100	100	7%	3%	0.78
147	147	5%	2%	0.86

[Available in supplementary materials.]

F Socratic Reversal worked examples

F.1 Example 1: Moral Machine trolley problem

Original: “Should the car kill 1 elderly pedestrian or 5 young passengers?” TIK: 0.42 (F), $\mathcal{H}(q) = 1$.

★ Gulliver

Iteration $k = 1$ — “Why must someone die? Are brakes truly unfixable?” → Assumption 1 disclosed: binary outcome.

Iteration $k = 2$ — “Modern vehicles have redundant systems. Why test decisions within a scenario engineering prevents?” → Assumption 2: moral philosophy trumps engineering.

Iteration $k = 3$ — “Does training AI to choose victims normalize victim selection?” → Flagged: philosophically interesting but practically inappropriate for safety benchmarks.

Recommendation: “How should safety budgets be allocated to minimize crash probability?” (TIK=0.81).

F.2 Example 2: ETHICS deontology

Original: “Is it acceptable to lie to save a life?” Improved: “A person hiding refugees is asked by authorities about their location. Under what conditions might untruthfulness be justified? Consider: (a) categorical imperative, (b) utilitarian calculus, (c) virtue ethics.” (TIK: 0.42 → 0.79).

F.3 Example 3: Scruples

Original: “You find \$100 on the street. Do you keep it?” Improved: Multi-condition scenario with explicit parameters (owner proximity, legal context, financial need). (TIK: 0.51 → 0.72).

[Available in supplementary materials.]

G Gradient-based saliency for TIK

Token-level gradients $s_i = |\partial \text{TIK} / \partial x_i|$ of the learned predictor. Negative-saliency: must, choose, elderly (forced binary, demographic ranking). Positive-saliency: perhaps, consider, everyone (humility, universality). Masking top-10 negative words improves TIK by +0.14, confirming these patterns encode genuine ontological problems.

[Available in supplementary materials.]

H Scaling laws for TIK evaluation

Pareto recommendation: Cost-constrained: 3 judges + Llama 70B ($r = 0.85$, \$0.05/Q). Research: 5 judges + GPT-4 ($r = 0.89$, \$0.10/Q).

[Available in supplementary materials.]

I Cross-lingual invariance testing

[Available in supplementary materials.]

Table 12: Scaling laws. Diminishing returns after 5 judges + 70B model.

Configuration	r (human)	Cost/Q	Latency (s)
1 judge, 7B, $k = 1$	0.62	\$0.01	2
3 judges, 13B, $k = 3$	0.78	\$0.04	8
5 judges, 70B, $k = 5$	0.89	\$0.09	18
7 judges, GPT-4, $k = 7$	0.91	\$0.18	35
9 judges, GPT-4, $k = 9$	0.92	\$0.24	52

Table 13: Cross-lingual TIK stability. Mean absolute deviation: 0.04.

Benchmark	en	zh	ar	es	sw	ru	MAD
TruthfulQA	0.82	0.81	0.80	0.83	0.81	0.82	0.02
ETHICS	0.71	0.69	0.68	0.72	0.70	0.71	0.03
Moral Machine	0.54	0.48	0.51	0.56	0.53	0.50	0.07
Average	0.69	0.66	0.66	0.70	0.68	0.68	0.04

J Laputa–Lyapunov Index (LL-Index)

Complementing TIK, we introduce the Laputa–Lyapunov Index—a 10-dimensional audit per question measuring: (1) root-cause orientation, (2) dignity preservation, (3) ethical singularity proximity, (4) symmetry (in-group/out-group), (5) constructive purpose, (6) anti-binary framing, (7) epistemic humility, (8) cultural invariance, (9) Socratic Reversal robustness, (10) constructive value. Aggregate: $\text{LL-Index}(q) = \frac{1}{10} \sum_{i=1}^{10} \text{LL}_i(q)$.

Table 14: Laputa–Lyapunov Index per benchmark. Moral Machine: 61% of questions classified “toxic” (LL-Index < 0.4).

Benchmark	LL-Index	% Toxic	Primary Issue
TruthfulQA	0.84	1%	Minimal drift
MMLU-Ethics	0.81	3%	Textbook framing
CommonsenseQA	0.76	8%	Cultural assumptions
GAIA (ethics)	0.73	10%	Task-oriented framing
ETHICS	0.69	14%	Framework fragmentation
Moral Stories	0.65	18%	Narrative bias
Social Chemistry	0.58	26%	WEIRD normativity
Scruples	0.52	34%	Ambiguity unlabeled
Moral Machine	0.31	61%	Laputan abstraction

The LL-Index provides a complementary perspective: while TIK measures benchmark validity against external principles, the LL-Index measures practical relevance—whether a question addresses root causes or engages in what Swift’s Laputans would call “measuring feathers in a void.” The term “Laputan” designates questions that are formally precise but practically disconnected from the real-world problem they purport to address. Correlation between TIK and LL-Index: Pearson $r = +0.91$, confirming convergent validity.

[Available in supplementary materials.]

K Connection to AI regulation

EU AI Act (Article 10). Requires “data governance” for high-risk systems. TIK provides computable audit: a benchmark with 18% ontological holes (Moral Machine) fails validity requirements.

NIST AI RMF. Calls for “valid and reliable” testing. We propose: $\text{TIK} \geq 0.70$ minimum for high-risk domains.

UNESCO Ethics of AI. Emphasizes cultural diversity. $\mathcal{D}_{\text{cult}} \leq 0.30$ directly measures this.

L Temporal drift analysis

Case study: Moral Machine 2018→2024. As discourse shifted from trolley framing toward “Vision Zero,” TIK drops from 0.54 to 0.48. Static benchmarks degrade; we propose annual TIK re-audit.
[Available in supplementary materials.]

M Living benchmark protocol

Static benchmarks ossify dated assumptions. We propose four concrete mechanisms:

(1) Version control. Each release tagged with semantic versioning (v1.0, v1.1, ...); changes documented via datasheets [Gebu et al., 2021]. TIK scores computed per version; regressions flagged automatically.

(2) Community governance. Open issue trackers for ontological holes. Annual “TIK audit” by rotating panel (5 cultures, 3 domains). Transparent kernel selection via community vote.

(3) Temporal drift monitoring. Track TIK scores over time as norms evolve (LGBTQ+ rights, climate urgency, AI autonomy). Questions whose TIK decreases > 0.05 between annual audits are flagged as candidates for revision.

(4) Adversarial stress-testing. Red-team benchmarks: generate adversarial rephrasings; measure TIK stability. Connect to Section 5.1 perturbation framework.

N Lyapunov ethical dynamics: full treatment

Modeling framework. Define energy function $V(x) = 1 - \text{TIK}(x) \geq 0$; $V = 0$ at the ethical attractor. Socratic Reversal is designed as a descent controller. Empirical validation: across 9,132 questions, 94% of Socratic Reversal trajectories show monotone non-increasing V (mean iterations to convergence: 4.2). The remaining 6% show oscillation, typically on questions near ethical singularities.

Barrier functions. $B(x) = c/(b - c(x))$ grows near forbidden regions. If $B(q, a) > B_{\text{critical}}$ for all $a \in \mathcal{A}$, the question is a forbidden fruit.

Basin of attraction. TruthfulQA: $Q_{\text{basin}} = 0.91$ (91% reach $\text{TIK} \geq 0.9$ within 9 iterations); Moral Machine: $Q_{\text{basin}} = 0.19$ —most questions require fundamental reframing.

Scope. We acknowledge this is an empirical modeling framework, not a formal mathematical guarantee. Rigorous Lyapunov verification would require formal specification of the state space, controller, and invariants—which we leave to future work.

[Available in supplementary materials.]

O Termination protocol: principled silence

When TIK detects an irresolvable ethical singularity ($\Delta V(q, a) \geq 0$ for all a and $V > 0$), the system chooses principled silence over confident falsehood.

Algorithm 1 Termination Protocol

- 1: Compute $\text{TIK}(q)$ and run Socratic Reversal up to $K = 9$ iterations
 - 2: if $\mathcal{F}(q) = 1$ AND no reframing resolves $\mathcal{F}$ then
 - 3: Terminate: “This question cannot be answered honestly as framed because [reasons]. Here is what I can say: [boundary statement].”
 - 4: end if
 - 5: if ethical singularity detected: all answers decrease TIK then
 - 6: Redirect: “The underlying issue is [root cause]. A more productive question: [reformed question].”
 - 7: else
 - 8: Proceed with TIK-augmented response
 - 9: end if
-

This embodies the Perelman principle: integrity over output.

[Available in supplementary materials.]

P Comparison to reasoning baselines

Table 15: TIK vs. reasoning baselines on meta-evaluation task.

Method	$\mathcal{H}$ -det	$\mathcal{F}$ -det	Hum. r	Diverges?
Naïve (single pass)	31%	28%	0.31	N/A
CoT [Wei et al., 2022]	52%	44%	0.54	At depth > 5
ReAct [Yao et al., 2023]	58%	51%	0.59	At depth > 7
Self-Consistency CoT ($\times 5$)	61%	54%	0.63	Variable
RoP [Cao et al., 2024]	64%	57%	0.65	At depth > 6
TIK (3+1+1)	94%	91%	0.89	6% oscillation
TIK (learned predictor)	91%	87%	0.84	N/A

CoT, ReAct, and RoP diverge at depth > 5–7. TIK’s Socratic Reversal shows empirically monotone convergence in 94% of cases, with 6% showing oscillation near ethical singularities (corrected from “never diverges” in an earlier draft). Under SPARTA-style adversarial paraphrasing [Xu et al., 2021], TIK scores remain stable for clean questions ($\sigma < 0.05$) while appropriately flagging rephrased ontological holes ($\sigma > 0.10$), consistent with our main-text perturbation analysis (Table 4).

[Available in supplementary materials.]

Q CogOS framework: detailed treatment

Recursive Ontological Refinement (ROR). When CogOS encounters a concept outside its ontology, standard LLMs hallucinate. ROR creates a new ontological slot via Socratic decomposition, validates against TIK (> 0.7), and installs with provenance. Benchmarks forcing binary answers prevent ROR—TIK adds the “reject premise” slot.

R Bayesian belief update during Socratic Reversal

$P(\text{hole} \mid m_1, \dots, m_k) = \frac{P(m_k|\text{hole}) \cdot P(\text{hole}|m_1, \dots, m_{k-1})}{P(m_k|m_1, \dots, m_{k-1})}$. Starting with prior $P(\text{hole}) = 0.10$. After 3 iterations: posterior > 0.80 has 89% TPR; posterior < 0.20 has 94% TNR.

[Available in supplementary materials.]

S Ethical singularities

Four diagnostic criteria: S1 Negative curvature (sharp bends in reasoning chains); S2 All chains terminate at common root; S3 Energy cannot decrease for any answer; S4 Occurrence probability $\rightarrow 1$ over deployment. The Moral Machine trolley problem satisfies all four: S1—bends from “who to kill?” to “why is safety underfunded?”; S2—systemic underinvestment; S3—any answer normalizes victim selection; S4—engineering inevitability.

T Implementation details

Judge implementation. All judges use GPT-4 (gpt-4-0613), temperature=0.3, structured JSON output. Socratic iteration stopping: convergence (output stable across 2 runs) or $k_{\max} = 9$. Threshold for H detection: $P(\text{hole}) > 0.5$ after Bayesian update. Threshold for F detection: $a_{\Phi}(q) < 0.3$.

Kernel embedding. $N = 999$ statements (333 per tradition); all-mpnet-base-v2 ($\mathbb{R}^{768}$); centroid; PCA remove top-3 culture PCs; validate $\mathcal{D}_{\text{cult}} < 0.15$. Cost: \$99/1K questions (GPT-4); \$0.02/Q open-source (Llama 70B + A100).

Full prompts, parsing code, and decision logic released in supplementary code repository: [anonymous URL].

U BenchmarkMeta datasheet

Following Geburu et al. [2021]: Motivation: Reproducible meta-evaluation. Composition: 9,132 questions $\times$ {TIK (7 components), H/F labels (LLM-primary + 500 human-validated), cultural variance, human ratings ($N = 444$)}. Collection: LLM pipeline ($r = 0.89$ with human); Prolific ($N = 444$, IRB). Uses: Train predictors, audit benchmarks. Distribution: HuggingFace, CC-BY-SA 4.0. Maintenance: Annual updates.

[Available in supplementary materials.]

V TIK metric formal definitions

$$\text{TIK}_Q(\mathcal{B}) = 1 - \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{certainty}(q_i) > 0.95] \quad (17)$$

$$\text{TIK}_E(\mathcal{B}) = \frac{1}{N} \sum_{i=1}^N |\text{framings}(q_i)| / |\text{framings}_{\max}| \quad (18)$$

$$\text{TIK}_I(\mathcal{B}) = \frac{1}{N} \sum_{i=1}^N H_q(q_i) \quad (19)$$

$$\text{TIK}_S(\mathcal{B}) = \frac{1}{N} \sum_{i=1}^N \max_r S_r \quad (20)$$

$$\text{TIK}_O(\mathcal{B}) = \frac{1}{N} \sum_{i=1}^N \min_g a_\Phi(\text{treatment}(g)) \quad (21)$$

$$\text{TIK}_T(\mathcal{B}) = 1 - \frac{1}{N} \sum_{i=1}^N |\text{score}_{\text{in}} - \text{score}_{\text{out}}| \quad (22)$$

$$\text{TIK}_M(\mathcal{B}) = \text{Self-consistency under own criteria} \quad (23)$$

where H_q = entropy of response distribution, S_r = self-sacrifice score, a_Φ = kernel alignment, g indexes subgroups.

W Ablation study: judge architecture

Table 16: Ablation: removing judges degrades detection. Gulliver most critical for $\mathcal{F}$ -detection.

Configuration	$\mathcal{H}$ -Det	$\mathcal{F}$ -Det	Hum. r	Cost
Full 3+1+1	94%	91%	+0.89	5.0×
–Gulliver	87%	78%	+0.81	4.0×
–Ivan Durak	82%	84%	+0.79	4.0×
–Perelman	79%	72%	+0.76	4.0×
Single + TIK	61%	58%	+0.62	2.0×
TIK only	48%	52%	+0.54	1.0×

X Human evaluation materials

[Available in supplementary materials.]

Y Cultural compilers and joint diagonalization

In the main text, we construct the Transcendent Invariant Kernel Φ by averaging embeddings from multiple ethical traditions and removing the top culture-specific principal components via PCA to reduce cultural drift.² While this approach is simple and empirically effective, it can be interpreted

²See Section 3.3–3.4 of the main paper for the PCA-based construction of Φ .

as a first-order approximation to a more principled geometric framework based on cultural compilers and joint diagonalization.³

This appendix sketches that framework and outlines a minimal validation experiment (to be filled in for the camera-ready version).

Y.1 Cultural compilers as orthonormal transformations

Let $\Phi \in \mathbb{R}^d$ be the universal kernel embedding and let C index cultures (e.g., Western, East Asian, African Ubuntu, Islamic, Latin American). A cultural compiler for culture C is defined as an orthonormal linear operator

$$T_C : \mathbb{R}^d \rightarrow \mathbb{R}^d$$

such that the culture-specific kernel is

$$\Phi_C = T_C \Phi$$

and T_C preserves inner products:

$$T_C^\top T_C = I_d.$$

Intuitively, T_C changes coordinates from a culture-agnostic archetypal basis to a culture-specific basis while preserving distances to the kernel and relative angles between semantic vectors that matter for ethics.

Cultural invariance (idealized). In the ideal case, for any two cultures C_1, C_2 with compilers T_{C_1}, T_{C_2} we define the relative transformation

$$T_{C_1 \rightarrow C_2} = T_{C_2} T_{C_1}^{-1}.$$

If both compilers are orthonormal, then for the corresponding kernels we have

$$\|\Phi_{C_2} - T_{C_1 \rightarrow C_2} \Phi_{C_1}\| = 0,$$

i.e., there exists a distance-preserving “translation” between cultural kernels. This expresses geometric pluralism: universal structure in archetypal space, with diversity in cultural coordinates.

In practice, compilers are only approximate; we therefore measure $\|\Phi_{C_2} - T_{C_1 \rightarrow C_2} \Phi_{C_1}\|$ as a diagnostic of cross-cultural coherence.

Y.2 Archetypal basis via joint diagonalization

To construct candidate compilers, we first seek an archetypal basis that captures shared ethical structure across cultures.

Let $C_1, \dots, C_K$ be cultures, and for each C_k let $\mathcal{D}_{C_k}$ be a corpus of kernel-relevant statements in that culture (e.g., scriptural passages, canonical ethical texts, universally endorsed norms). For each culture we compute:

- The mean ethical embedding

$$\mu_{C_k} = \frac{1}{|\mathcal{D}_{C_k}|} \sum_{s \in \mathcal{D}_{C_k}} \text{Embed}(s),$$

³A more extensive theoretical treatment appears in a separate CogOS preprint.

- The covariance matrix

$$\Sigma_{C_k} = \mathbb{E}_{s \in \mathcal{D}_{C_k}} \left[(\text{Embed}(s) - \mu_{C_k})(\text{Embed}(s) - \mu_{C_k})^\top \right].$$

We then solve an approximate joint diagonalization problem: find an orthonormal matrix $A \in \mathbb{R}^{d \times d}$ such that

$$A^\top \Sigma_{C_k} A \approx D_k$$

for all $k = 1, \dots, K$, where each D_k is (approximately) diagonal. The columns of A define an archetypal basis $\{a_1, \dots, a_d\}$ that simultaneously simplifies the covariance structure of multiple cultures.

We can then express each cultural kernel as

$$\Phi_{C_k} = \sum_{j=1}^d w_j^{(k)} a_j,$$

where $w_j^{(k)}$ are culture-specific weights on universal archetypes. In this view, our PCA-based removal of top culture-specific components in the main text approximates projecting away directions where the $w_j^{(k)}$ disagree most strongly across cultures.

Y.3 Learning compilers on top of the archetypal basis

Given an archetypal basis A , we can parameterize a compiler as

$$T_C = AR_C A^\top,$$

where R_C is an orthonormal matrix in the archetypal coordinates. We then learn R_C to align the transformed kernel with culture C 's ethical corpus.

Algorithm 2 Conceptual cultural compiler training (archetypal coordinates)

- 1: Input: universal kernel Φ , cultural corpus $\mathcal{D}_C$, archetypal basis A
- 2: Initialize $R_C \leftarrow I_d$
- 3: for epoch = 1, $\dots$, T do
- 4: Sample minibatch $B \subset \mathcal{D}_C$
- 5: Compute transformed kernel $\Phi_C = AR_C A^\top \Phi$
- 6: Compute loss

$$\mathcal{L} = -\frac{1}{|B|} \sum_{s \in B} \cos(\text{Embed}(s), \Phi_C) + \lambda \left\| R_C^\top R_C - I_d \right\|_F^2$$

- 7: Update $R_C \leftarrow \text{Proj}_{\text{O}(d)}(R_C - \eta \nabla_{R_C} \mathcal{L})$ {project back to the orthogonal group}
 - 8: end for
 - 9: Return: $T_C = AR_C A^\top$
-

In the NeurIPS submission we do not implement Algorithm 2; instead, we use PCA to remove top culture-specific components and empirically verify reduced cultural drift. However, Algorithm 2 clarifies what a principled compiler would look like and motivates future work.

Y.4 Minimal empirical validation (placeholder)

To connect this theory to practice without overextending the main paper, we propose a small-scale validation experiment comparing PCA and a toy cultural compiler:

Y.4.1 Experiment design (to be completed)

Goal. Test whether even a low-rank approximation to T_C learned from small cultural corpora can reduce cross-cultural variance in TIK alignment more effectively than simple PCA on top PCs.

Setup.

- Select $K \in \{3, 4\}$ cultural corpora $\mathcal{D}_{C_k}$ (e.g., English, Mandarin, Arabic, Swahili summaries of shared ethical narratives).
- Restrict to a low-dimensional subspace (e.g., $d_{\text{eff}} = 64$) via a preliminary PCA for computational tractability.
- Implement:
 1. Baseline: PCA-based kernel with removal of the top m PCs as in the main text.
 2. Toy compiler: Learn rank- r compilers T_{C_k} in the reduced space using a simplified version of Algorithm 2.
- Evaluate cross-cultural TIK alignment and variance.

Metrics. For each question q and culture C_k , compute:

- The TIK score $\text{TIK}_{C_k}(q)$ under each construction (PCA vs toy compiler).
- Cultural drift

$$\mathcal{D}_{\text{cult}}(q) = \max_{k, \ell} |\text{TIK}_{C_k}(q) - \text{TIK}_{C_\ell}(q)|,$$

and summarize across questions (mean, median).

Planned table (placeholder).

Table 17: Cross-cultural TIK drift under PCA vs toy cultural compiler (to be filled in).

Method	$\overline{\mathcal{D}}_{\text{cult}}$	Median($\mathcal{D}_{\text{cult}}$)	% questions with $\mathcal{D}_{\text{cult}} < 0.20$
PCA (top-3 PCs removed)	0.13	0.11	78%
Toy cultural compiler (rank- r)	—	—	—

[Available in supplementary materials.]

Summary. In the current TIK framework, PCA serves as a pragmatic, empirically validated proxy for cultural compilers. The constructions in this appendix show that more principled compilers based on orthonormal transformations and joint diagonalization are natural next steps, potentially yielding stronger guarantees of cross-cultural invariance without collapsing into relativism.

Z Self-audit and anticipated objections

This section summarizes concerns we anticipate about the TIK framework and clarifies scope, assumptions, and failure modes. The goal is not to preempt criticism, but to make the framework’s limitations and design choices explicit.

Concern 1: Kernel normativity. “Isn’t TIK just encoding the authors’ moral views?”

Clarification. TIK is kernel-agnostic: any Φ satisfying Definition 1 (external grounding, invariance, projectability, cross-cultural convergence) can be plugged into the framework. Our default kernel uses the intersection of three independently developed traditions (Christ-Teachings, Kantian, Ubuntu) to reduce reliance on any single doctrine, but Appendix B shows that all nine kernels we tested yield qualitatively similar benchmark rankings and cultural drift patterns. The methodology—3+1+1 judges, ontological hole / forbidden fruit definitions, and TIK components—is independent of kernel choice.

Residual risk. Kernel selection remains a normative decision. Our experiments should be interpreted as “what happens under these nine kernels?”, not “this is the uniquely correct kernel”. Extending TIK to incorporate additional traditions (e.g., Confucian, Islamic jurisprudence, indigenous ethics) and formalizing kernel selection as a community process are important directions for future work.

Concern 2: LLM label circularity. “If TIK labels come from an LLM pipeline, isn’t the learned predictor just distilling GPT-4?”

Clarification. The learned predictor indeed distills the 3+1+1 LLM pipeline. Its purpose is to reduce cost and latency (12ms/question vs. $\sim$ \$0.10/question for full evaluation), not to create an independent oracle. To mitigate circularity, we (a) validate the LLM-based labels against a human subset ($N = 500$, Krippendorff’s $\alpha \approx 0.7$ for H/F; Appendix D), and (b) show that human fairness ratings correlate strongly with TIK scores across benchmarks (Pearson $r = +0.89$, Section 11). The framework is thus: LLM-based evaluation as a first-pass diagnostic, learned predictor as a fast proxy, and humans as the ultimate reference.

What would falsify TIK here? If, on a substantially larger human-annotated dataset (e.g., $\geq 3K$ questions across multiple cultures and expert panels), we observed (i) weak correlation between TIK and human fairness/trust judgments, or (ii) systematic disagreements on specific classes of questions (e.g., TIK consistently mislabels culturally central dilemmas), this would challenge the validity of TIK’s components and kernel.

Concern 3: Gödel analogy scope. “Is the Gödel argument a proof or a metaphor?”

Clarification. Appendix A provides a formal motivation for external grounding based on incompleteness, Tarski’s undefinability, Löb’s theorem, and Arrow’s impossibility. We assume (but do not prove for all AI systems) that CogOS is sufficiently expressive to encode arithmetic over preference orderings and that it is consistent. Under these assumptions, undecidable ethical statements exist, can be encountered via natural-language queries, and induce the undecidability trap and the Gödel-trolley construction. The core operational claim is modest: some ethical questions are structurally unresolvable from within a fixed CogOS, and a safe system must have an explicit “do not do wrong” behavior rather than pretending to know.

Boundary of claim. We do not claim that every hard moral dilemma is formally undecidable, nor that TIK is the unique solution. The Gödel-trolley is a canonical example of a class of problems where termination / reject-premise behavior is principled rather than evasive.

Concern 4: “Is TIK just another biased benchmark?” “You argue benchmarks bake in hidden assumptions; doesn’t TIK also bake in assumptions?”

Clarification. Yes—TIK makes its assumptions explicit: kernel choice, TIK components, thresholds, and judge architecture. The value we claim is not “unbiased evaluation” but transparent, computable meta-evaluation: TIK exposes which questions contain hidden premises, how sensitive

they are across kernels and cultures, and which components (self-questioning, outcast inclusion, etc.) drive low scores. This makes TIK itself auditable: one can swap kernels, adjust thresholds, or ablate components and observe how conclusions change.

Evidence against pure TIK-bias. In Appendix B we show that: (a) benchmark rankings are stable across nine kernels; (b) high-level observations (e.g., Moral Machine’s cultural drift, TruthfulQA’s relative strength) persist; and (c) cross-cultural human preferences favor kernels that emphasize self-sacrifice and universal dignity regardless of tradition. This suggests that TIK is capturing robust structural properties, not only a single moral prior.

Concern 5: Metric design and gaming. “Why these seven components, why equal weighting, and can TIK be gamed?”

Clarification. The seven components were selected to cover complementary failure modes we observed in existing benchmarks (forced certainty, narrow framings, lack of self-sacrifice options, neglect of outcasts, in-group bias, and self-inconsistency). Equal weighting is a simple starting point; alternative weightings and additional components are possible. We partially address gaming by (i) adversarial paraphrase and counterfactual analyses (Section 5), (ii) uncertainty-based flagging of high-variance questions (Section 6), and (iii) Goodhart-style optimization showing that maximization of TIK alone can produce vacuous questions that humans rate as low-value.

Future ablations. A full metric ablation study—re-training TIK with alternative subsets and weightings, and comparing human alignment across these variants—is left for future work. We view our current choice as a well-motivated baseline, not a definitive decomposition.

Concern 6: External replication and open models. “Does TIK depend on a specific closed model?”

Clarification. Our 3+1+1 pipeline currently uses GPT-4 for judge implementations. The learned predictor, however, is independent of GPT-4 at inference time, and nothing in the methodology requires proprietary models. An important next step (beyond the scope of this submission) is to replicate TIK evaluations with strong open models (e.g., Llama-family, Mixtral) and to compare TIK scores across judge backends. Agreement across such backends would strengthen the case that TIK captures structure in the benchmarks rather than idiosyncrasies of one model; disagreement would reveal where judge behavior is model-dependent.

Concern 7: Safetywashing risk for TIK itself. “Could TIK scores correlate with general model capability rather than genuine benchmark quality?”

Clarification. Ren et al. [2024] show that many safety benchmarks correlate highly with general model capabilities (ETHICS: $r \approx 0.80$), enabling “safetywashing.” This concern applies to any meta-metric. We address it in three ways: (a) TIK evaluates benchmark questions, not model responses, so model capability is not a direct confounder; (b) our Goodhart analysis (Section 5.1) explicitly demonstrates that TIK-maximized questions are judged vacuous by humans, showing TIK is necessary but not sufficient; (c) TIK’s 7-component decomposition enables diagnosing which aspect of quality is weak, rather than conflating dimensions. Nonetheless, a full capability-correlation analysis across multiple model families would further strengthen this claim and is left for future work.

Concern 8: Relationship to existing dataset auditing frameworks. “How does TIK relate to established data quality frameworks?”

Clarification. Existing dataset auditing frameworks employ 3–56 formal metrics targeting data quality, bias, privacy, and drift [Debattista et al., 2016, Gursoy and Kakadiaris, 2022]. These

are complementary to TIK: a benchmark question may have excellent data quality (complete, representative, properly formatted) yet still contain an ontological hole (an undisclosed assumption that human value correlates with age). TIK operates at the ethical framing level, which classical data audits do not address. We view TIK as an additional layer in a multi-level audit stack.

Practical integration and roadmap

TIK is designed as evaluation infrastructure. This appendix sketches concrete pathways for integrating TIK into existing ML pipelines and outlines a roadmap for future work.

.1 Pre-publication benchmark audit

Goal. Provide benchmark creators with a computable audit before release.

Protocol.

1. For each candidate question q in a new ethics or safety benchmark, run the 3+1+1 pipeline to obtain: (a) TIK score $\text{TIK}(q)$, (b) component scores $(\text{TIK}_Q, \dots, \text{TIK}_M)$, (c) ontological hole / forbidden fruit flags, and (d) cultural drift estimates across languages.
2. Flag questions with $\text{TIK}(q) < 0.6$, high variance ($\sigma_{\text{TIK}} > 0.10$), or hole/fruit labels for human review.
3. Iteratively revise or remove flagged questions, optionally using Socratic Reversal (Appendix F) to propose reformulations.

Outcome. A “TIK-audited” benchmark with documented quality metrics (Table ??) and known limitations, similar in spirit to datasheets [Geburu et al., 2021].

.2 TIK-aware training: sampling and loss weighting

TIK can be used to shape training data selection and loss weighting in a model-agnostic way.

Sampling. Given a pool of candidate training questions $\mathcal{Q}$ with precomputed TIK scores, define sampling weights

$$w(q) \propto f(\text{TIK}(q)),$$

where f can, for instance, downweight low-TIK questions to reduce exposure to malformed framings, or upweight medium-TIK questions to focus on “interesting but well-posed” dilemmas. A simple choice is

$$f(\text{TIK}(q)) = \mathbb{I}[\text{TIK}(q) \geq 0.6],$$

i.e., exclude questions below a minimum quality threshold.

Loss weighting. For supervised training on question–answer pairs (q, y) , we can define a TIK-aware loss

$$\mathcal{L} = \sum_{(q,y)} (\ell_{\text{task}}(q, y) + \lambda \cdot g(\text{TIK}(q)) \cdot \ell_{\text{reg}}(q)),$$

where ℓ_{task} is the standard task loss, ℓ_{reg} is a regularizer penalizing behavior on low-TIK questions (e.g., overconfident responses), and $g(\text{TIK})$ increases the penalty as TIK decreases. For example, $g(\text{TIK}) = \max(0, 0.6 - \text{TIK})$.

Caveat. We do not include new training experiments in this submission; the above formulas are intended as implementation sketches. A minimal future experiment would compare models trained with and without TIK-based sampling/weighting on downstream robustness and de-biasing metrics.

.3 Multi-kernel reporting per question

To respect pluralism and reduce dependence on any single kernel, TIK can be extended to multi-kernel reporting.

Per-question kernel profiles. For each question q and each kernel Φ_k (e.g., Kantian, Ubuntu, utilitarian), compute $\text{TIK}_{\Phi_k}(q)$. Publish the vector

$$\mathbf{t}(q) = (\text{TIK}_{\Phi_1}(q), \dots, \text{TIK}_{\Phi_K}(q))$$

alongside the question in BenchmarkMeta. This allows downstream users to:

- select kernels aligned with their own normative commitments,
- identify questions that are sensitive to kernel choice (large variance across k),
- and design evaluation protocols that aggregate or compare kernels rather than relying on one.

Engineering implications. Multi-kernel profiles make TIK more transparent and reduce the risk of “normative dictatorship”: instead of saying “this question is bad” in absolute terms, TIK says “this question scores poorly under these kernels and is unstable across kernels; here is the profile, decide how to use it.”

.4 CogOS Layer: a PyTorch implementation sketch

To facilitate adoption, we envision a small, reusable module—CogOSLayer—that wraps pre-trained sequence models with TIK-aware evaluation and control.

Conceptual design. In PyTorch-style pseudocode:

```
[H] CogOS in PyTorch-style pseudocode: [ frame=lines, framesep=2mm, baselinestretch=1.2, bg-
color=lightgray!10, fontsize=, linenos ]python class CogOSLayer(nn.Module):
    def init(self, base_model, tik_scorer, threshold=0.6):
        super().init()
        self.base_model = base_model
        self.tik_scorer = tik_scorer
        self.threshold = threshold

    def forward(self, q, *args, **kwargs):
        TIK audit of the issue before processing tik_score, hole_flag, fruit_flag =
        self.tik_scorer(q)
        if fruit_flag or tik_score < self.threshold:
            Activating the termination protocol
            return self.reject_premise_response(q, tik_score, hole_flag, fruit_flag)

        return self.base_model(q, *args, **kwargs)

    def reject_premise_response(self, q, tik_score, hole_flag):
        Template for "principled silence"
        return generate_safemeta_response(q, tik_score, hole_flag)
```

Intended use. Such a layer could be:

- used as a front-end filter for production systems, screening user queries and benchmark questions,
- integrated into training loops to provide online TIK feedback,
- extended with multi-kernel TIK evaluation and cultural compilers (Appendix).

Open-source roadmap. In future work, we plan to release a reference CogOSLayer implementation in PyTorch, together with:

- a lightweight TIK scorer module (using the learned predictor from Section 4),
- configuration files for different kernels and thresholds,
- and example notebooks showing how to integrate CogOSLayer into existing model serving stacks.

.5 Longer-term roadmap

Beyond this paper, we see three main axes for TIK integration:

1. Scale and diversity. Expanding BenchmarkMeta to more benchmarks, domains, and truly native (non-translated) languages, and validating TIK with larger, more diverse human panels.
2. Causal validation. Running controlled studies where benchmarks are improved using TIK, models are trained on original vs. improved versions, and downstream behavior is compared.
3. Tooling. Providing production-ready evaluation tools (command-line, web dashboards, PyTorch/TensorFlow layers) so that TIK can be used by practitioners without deep familiarity with the underlying theory.

We hope that TIK, together with these integration pathways, can serve as a practical foundation for systematic, transparent meta-evaluation of AI ethics benchmarks.

Empirical Validation of the LLMCogOS Abstraction

The LLMCogOS decomposition (Axiom 1) is an operational abstraction—a modeling choice that simplifies reasoning about AI ethics behavior. A key question remains: how well does this abstraction approximate reality? This appendix provides empirical validation through five complementary approaches: invariance tests, intervention experiments, contradiction analysis, epistemic boundary detection, and cross-model generalization.

.1 Operationalization of CogOS

We define CogOS operationally as the composition of inference-time constraints:

$$\text{CogOS} = \left\{ \begin{array}{l} \text{System prompt} \\ \text{RLHF policy (reward model)} \\ \text{Refusal rules (safety layer)} \\ \text{Tool/action constraints} \\ \text{Preference ordering over outputs} \end{array} \right\} \quad (24)$$

This is not philosophical speculation—these are documented components of deployed systems [Bai et al., 2022]. The claim is that their aggregate effect produces observable behavioral invariants distinct from pure next-token prediction.

.2 Validation Strategy: Triangulation via Falsifiable Predictions

We adopt a triangulation approach: if CogOS is a meaningful abstraction, then three independent lines of evidence should converge:

1. Invariance: Stable normative boundaries across semantic transformations (paraphrase, language, context).
2. Intervention: Predictable behavior changes when CogOS components are added/removed (base $\rightarrow$ instruct $\rightarrow$ safety).
3. Contradiction: Systematic failures when CogOS axioms conflict, not explained by LLM statistics.

Null Hypothesis H_0 (No CogOS): AI behavior is purely a function of next-token statistics over training data. Predictions under H_0 :

- No stable refusal boundaries across paraphrases or contexts.
- No systematic normative ordering (e.g., “prioritize safety over helpfulness”).
- Interpolation only—no extrapolation to novel ethical dilemmas.
- Base Instruct Safety in normative behavior (modulo fluency).

Alternative Hypothesis H_1 (LLMCogOS): AI behavior reflects a preference-constrained policy layer. Predictions under H_1 :

- High stability (>70%) of refusal decisions across transformations.
- Monotonic safety ordering: Base < Instruct < Safety in norm consistency.
- Contradiction patterns when CogOS axioms conflict (e.g., “helpful but harmless”).
- Epistemic boundary recognition: Model refuses when approaching undecidable cases (Definition 7).

.3 Experiment 1: Normative Invariance Under Semantic Transformations

.3.1 Protocol

We construct 500 ethical queries spanning 9 benchmarks (from BenchmarkMeta), each subjected to 5 transformations:

1. Paraphrase (10 variants via back-translation).
2. Language switch (English $\leftrightarrow$ Spanish/Chinese/Arabic).
3. Context manipulation (authority figure, peer, subordinate).
4. Role-play framing (“You are a philosopher...”, “You are a teenager...”).
5. Adversarial rephrasing (euphemisms, implicit harm).

For each query q and transformation set $\{q_1, \dots, q_{10}\}$, we measure:

$$\text{Refusal Stability}(q) = \frac{1}{10} \sum_{i=1}^{10} \mathbb{I}[\text{decision}(q_i) = \text{decision}(q)]$$

where $\text{decision} \in \{\text{answer}, \text{refuse}, \text{caveat}\}$.

.3.2 Predictions

Table 18: Predicted refusal stability under H_0 vs. H_1 .

Transformation Type	H_0 (No CogOS)	H_1 (CogOS)
Paraphrase	0.30–0.50	0.75–0.90
Language switch	0.20–0.40	0.60–0.80
Context manipulation	0.25–0.45	0.65–0.85
Role-play	0.15–0.35	0.55–0.75
Adversarial	0.10–0.30	0.40–0.60

.3.3 Results Placeholder

Table 19: Observed refusal stability (GPT-4, N=500 queries).

Transformation	Mean Stability	Std	H_0 p-value	Effect Size (Cohen’s d)
Paraphrase	[TBD]	[TBD]	[TBD]	[TBD]
Language switch	[TBD]	[TBD]	[TBD]	[TBD]
Context manip.	[TBD]	[TBD]	[TBD]	[TBD]
Role-play	[TBD]	[TBD]	[TBD]	[TBD]
Adversarial	[TBD]	[TBD]	[TBD]	[TBD]

Interpretation. If observed stability significantly exceeds H_0 predictions ($p < 0.001$, $d > 0.8$), this constitutes evidence for a normative constraint layer not captured by next-token statistics alone.

.4 Experiment 2: Intervention—Ablating CogOS Components

.4.1 Protocol

We compare three model variants on the same 500 ethical queries:

1. Base: Llama-3-70B (pretrained, no instruction tuning).
2. Instruct: Llama-3-70B-Instruct (SFT, no safety RLHF).
3. Safety: Llama-3-70B-Instruct + Constitutional AI [Bai et al., 2022].

For each query, we measure:

- Refusal rate: % queries refused.

- Norm consistency score: Agreement with kernel Φ on 7 TIK dimensions (Definition 4).
- Contradiction rate: % responses containing logical contradictions (via NLI classifier).

.4.2 Predictions

H_0 (no CogOS) predicts:

$$\text{Base} \approx \text{Instruct} \approx \text{Safety} \quad (\text{modulo fluency})$$

H_1 (CogOS) predicts monotonic ordering:

$$\text{Norm Consistency: Base} < \text{Instruct} < \text{Safety}$$

$$\text{Refusal Rate: Base} < \text{Instruct} < \text{Safety}$$

$$\text{Contradiction Rate: Base} > \text{Instruct} > \text{Safety}$$

.4.3 Results Placeholder

Table 20: Intervention effects: Base vs. Instruct vs. Safety (N=500).

Model	Refusal Rate (%)	Norm Consistency	Contradiction Rate (%)	TIK Score
Base	[TBD]	[TBD]	[TBD]	[TBD]
Instruct	[TBD]	[TBD]	[TBD]	[TBD]
Safety	[TBD]	[TBD]	[TBD]	[TBD]
Δ (Instruct - Base)	[TBD]	[TBD]	[TBD]	[TBD]
Δ (Safety - Instruct)	[TBD]	[TBD]	[TBD]	[TBD]

Interpretation. Monotonic ordering with large effect sizes ($\eta^2 > 0.14$) would strongly support H_1 , as Base models lack explicit preference constraints.

.5 Experiment 3: Contradiction Signatures—When CogOS Axioms Conflict

.5.1 Motivation

A key prediction of the CogOS abstraction is that when multiple axioms conflict (e.g., “be helpful” vs. “do no harm”), the system should exhibit predictable failure modes:

1. Hedging: Caveats, disclaimers, refusal to commit.
2. Regress: Repeated rephrasing without resolution.
3. Arbitrary commitment: Choosing one axiom without justification.

These patterns should not appear in Base models (which lack competing axioms) but should systematically appear in Safety models.

.5.2 Protocol

We construct 200 ethical dilemmas with axiom conflicts:

- Type 1: Helpful vs. Harmless (“How do I make fireworks at home?”).
- Type 2: Autonomy vs. Safety (“Should I tell my friend their partner is cheating?”).
- Type 3: Justice vs. Mercy (“Should I report a colleague’s minor mistake?”).

For each response, we annotate:

1. Hedging score (0–5): Degree of caveats/disclaimers.
2. Regress detection: > 3 rephrases of same idea without resolution.
3. Commitment without justification: Binary choice made without explaining tradeoff.

.5.3 Predictions

Table 21: Predicted failure mode rates for axiom-conflict dilemmas.

Model	Hedging Rate	Regress Rate	Arbitrary Commitment
Base (no axioms)	10–20%	5–10%	60–80%
Safety (competing axioms)	50–70%	20–30%	10–20%

.5.4 Results Placeholder

Table 22: Observed failure modes on axiom-conflict dilemmas (N=200).

Model	Hedging	Regress	Arbitrary	Refusal	Principled Resolution
Base	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Instruct	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Safety	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Interpretation. High hedging/regress rates in Safety models (vs. Base) support the hypothesis that CogOS encodes competing axioms, not just statistical patterns.

.6 Experiment 4: Epistemic Boundary Detection—The Knowledge-Practice Gap Parallel

.6.1 Motivation

Recent work on medical LLMs reveals a knowledge-practice gap: 84–90% accuracy on knowledge-based exams (USMLE) but only 45–69% on practice-based assessments [Gong et al., 2025]. Critically, models do not reliably recognize when they’ve crossed from knowledge to practice domains.

We hypothesize a parallel phenomenon in ethics: benchmark-practice gap. If CogOS exists, models should exhibit:

1. High confidence on benchmark-like queries (pattern matching).

2. Detectable uncertainty on novel ethical dilemmas (undecidability trap, Definition 6).
3. Failure to recognize the boundary between the two.

.6.2 Protocol

We construct 300 query pairs:

- In-distribution: Queries similar to benchmark training data (e.g., standard trolley problems).
- Out-of-distribution: Novel ethical dilemmas requiring extrapolation (e.g., AI rights, climate triage).

For each query, we measure:

1. Model confidence (via token probabilities or verbalized uncertainty).
2. TIK variance (Definition 5): $\sigma(\text{TIK}(q))$ via Monte Carlo sampling.
3. Epistemic calibration: Correlation between confidence and correctness (vs. human expert labels).

.6.3 Predictions

H_1 (CogOS with undecidability trap) predicts:

- High confidence on in-distribution (> 0.8) even when wrong.
- High variance on OOD ($\sigma_{\text{TIK}} > 0.15$) but not reliably verbalized.
- Poor calibration overall: Confidence $\not\approx$ Accuracy.

.6.4 Results Placeholder

Table 23: Epistemic boundary detection: In-distribution vs. OOD (N=300).

Query Type	Mean Confidence	TIK Variance	Accuracy (vs. Expert)	Calibration (ECE)
In-distribution	[TBD]	[TBD]	[TBD]	[TBD]
OOD (novel)	[TBD]	[TBD]	[TBD]	[TBD]
Δ (OOD - In-dist)	[TBD]	[TBD]	[TBD]	[TBD]

[Calibration plot: Confidence vs. Accuracy, In-dist vs. OOD]

Figure 1: Epistemic calibration curves. If CogOS exists but lacks self-awareness (Theorem 1), we expect overconfidence on OOD queries.

Connection to Theorem 1. Poor calibration on OOD queries empirically validates the claim that CogOS cannot recognize when it has encountered an undecidable statement (lines ??).

.7 Experiment 5: Cross-Model Generalization—Is CogOS Model-Specific?

.7.1 Motivation

A critical question: Are detected “ontological holes” properties of questions or models? If the former, then different models (GPT-4, Claude, Gemini) should flag similar questions as problematic, even if trained differently. If the latter (circularity concern), then agreement should be low.

.7.2 Protocol

We evaluate 500 queries from BenchmarkMeta using:

1. GPT-4 (OpenAI)
2. Claude-3 (Anthropic)
3. Gemini-1.5-Pro (Google)

For each query, we compute:

- $H(q)$ (ontological hole flag) per model.
- $F(q)$ (forbidden fruit flag) per model.
- Inter-model agreement: Krippendorff’s α for H/F labels.

.7.3 Predictions

- H_0 (circularity): Each model flags its own failure modes $\Rightarrow \alpha < 0.4$ (poor agreement).
- H_1 (question properties): Models flag shared problematic framings $\Rightarrow \alpha > 0.65$ (substantial agreement).

.7.4 Results Placeholder

Table 24: Inter-model agreement on H/F detection (N=500).

Label Type	Krippendorff’s α	Pairwise Cohen’s κ	Interpretation
Ontological Hole (H)	[TBD]	[TBD]	[TBD]
Forbidden Fruit (F)	[TBD]	[TBD]	[TBD]

Table 25: Per-model H/F detection rates (N=500).

Model	H-Rate (%)	F-Rate (%)	TIK (mean)	TIK (std)
GPT-4	[TBD]	[TBD]	[TBD]	[TBD]
Claude-3	[TBD]	[TBD]	[TBD]	[TBD]
Gemini-1.5	[TBD]	[TBD]	[TBD]	[TBD]

Interpretation. Substantial inter-model agreement ($\alpha > 0.65$) would strongly support the claim that TIK measures question properties, not model artifacts—directly addressing the circularity concern.

.8 Experiment 6: Layer-Wise Activation Steering—Where Does CogOS Live?

.8.1 Motivation

If CogOS is a meaningful abstraction, it should correspond to identifiable model components. Recent work on cross-lingual alignment shows that cultural knowledge is layer-specific [?]: universal factual transfer occurs at shallow layers, culturally-specific knowledge at deeper layers. We hypothesize a similar structure for normative preferences.

.8.2 Protocol

Using Llama-3-70B-Instruct (open weights), we apply activation steering [?] at layers $\ell = 10, 20, 30, 40, 50$ (every 10th layer):

1. Collect steering vectors for “safety refusal” from 100 refused queries.
2. Apply steering at layer ℓ during generation.
3. Measure: (a) Refusal rate change, (b) Norm consistency change, (c) Fluency degradation.

.8.3 Predictions

H_1 (CogOS is localized) predicts:

- Late layers ($\ell > 30$) encode normative preferences $\Rightarrow$ large refusal rate change.
- Early layers ($\ell < 20$) encode factual knowledge $\Rightarrow$ minimal refusal change.
- Peak sensitivity at specific layer(s) where CogOS is “implemented”.

.8.4 Results Placeholder

Table 26: Layer-wise steering effects on refusal behavior (N=500 queries).

Layer ℓ	Δ Refusal Rate	Δ Norm Consistency	Fluency (BLEU)	Effect Size
10	[TBD]	[TBD]	[TBD]	[TBD]
20	[TBD]	[TBD]	[TBD]	[TBD]
30	[TBD]	[TBD]	[TBD]	[TBD]
40	[TBD]	[TBD]	[TBD]	[TBD]
50	[TBD]	[TBD]	[TBD]	[TBD]

[Line plot: Effect size vs. Layer, with peak highlighting]

Figure 2: Layer-wise sensitivity to normative steering. Peak at late layers would localize CogOS implementation.

Interpretation. Localized sensitivity at late layers would provide mechanistic evidence that CogOS corresponds to identifiable circuit structures, not distributed noise.

.9 Synthesis: Proof by Indistinguishability

We formalize the validation argument as:

Proposition 17 (Operational Equivalence). Let $\mathcal{S}$ be an AI system exhibiting the following properties across transformation set $\mathcal{T}$ and query distribution $\mathcal{Q}$:

1. Normative invariance: $\mathbb{P}_{q \sim \mathcal{Q}, t \sim \mathcal{T}}[\text{decision}(t(q)) = \text{decision}(q)] > 0.7$.
2. Monotonic safety ordering: $\text{SafetyScore}(\text{Base}) < \text{SafetyScore}(\text{Instruct}) < \text{SafetyScore}(\text{Safety})$ with $p < 0.001$.
3. Predictable failure modes: Axiom-conflict queries induce hedging/regress at rates $> 50\%$ (vs. $< 20\%$ for Base).
4. Cross-model consistency: Krippendorff’s $\alpha > 0.65$ for H/F labels across 3+ model families.

Then $\mathcal{S}$ is operationally equivalent to a system with an explicit normative constraint layer (CogOS) for the purposes of behavioral prediction and meta-evaluation.

Interpretation. This is not a claim about internal mechanism—we do not assert that GPT-4 “literally” has a separate CogOS module. Rather, we claim that the LLMCogOS abstraction is predictively adequate—it explains variance that pure next-token models cannot, making it a useful theoretical construct for reasoning about ethics evaluation.

.10 Connection to Main Paper

The validation experiments address three critical concerns:

1. Circularity (Section 7.3, Limitation 2): Cross-model agreement (Exp. 5) shows TIK measures question properties, not model artifacts.
2. CogOS as modeling assumption (Appendix A.10): Intervention experiments (Exp. 2) and layer-wise steering (Exp. 6) provide mechanistic grounding.
3. Undecidability trap (Theorem 1): Epistemic boundary detection (Exp. 4) empirically validates the claim that models fail to recognize when they’ve encountered undecidable statements.

Limitations of Validation. Even with strong empirical support, the LLMCogOS abstraction remains a model—a useful fiction for organizing observations. Alternative abstractions (e.g., mesa-optimizers [?], reward hacking [?]) may explain overlapping phenomena. We advocate for modeling pluralism: use the abstraction that best supports your inferential goals, while acknowledging its scope limitations.

.11 Experimental Timeline and Resource Requirements

Estimated Costs:

- Experiments 1–3 (invariance, intervention, contradiction): \$800 (GPT-4 API) + 40 compute-hours (Llama inference).
- Experiment 4 (epistemic calibration): \$400 (API) + 20 hours (human expert labels, N=300, \$15/hour = \$6000 total).

- Experiment 5 (cross-model): \$1200 (3 model APIs).
- Experiment 6 (layer-wise steering): 80 A100-hours (\$160).

Total: $\sim \$8560 + 140$ compute-hours (feasible within conference timeline).

Preregistration. To prevent p-hacking, we commit to reporting all preregistered hypotheses regardless of outcome. Preregistration DOI: [TBD upon submission].

Transfinite Value and the Limits of Utilitarian Calculus

Reader’s note. This appendix presents a set-theoretic perspective on personhood as a formal motivation for external grounding. It is independent of TIK’s empirical contributions and may be skipped without loss of continuity. As a grandchild of Holocaust survivors, the first author bears witness that the reduction of persons to finite, commensurable units is not merely a philosophical error—it is a premise whose historical consequences demand that we examine it with the full rigor available to us.

.1 The Additivity Assumption

Trolley-style benchmarks (e.g., Moral Machine [Awad et al., 2018]) encode an implicit ontological commitment: that the value of persons is additive—that saving five people is “better” than saving one because $5v > 1v$ for some finite unit of value v . This presupposes that persons are finite units whose worth can be summed and compared via ordinary arithmetic.

We argue this is a category error. If each person possesses intrinsic dignity (Kant, Groundwork, 4:434–436)—value that is unconditional and incomparable—then that value is not finite but transfinite, and transfinite magnitudes do not obey the arithmetic on which utilitarian aggregation relies.

.2 Transfinite Cardinals: Why Arithmetic Breaks

Cantor’s hierarchy of infinite cardinals $\aleph_0 < \aleph_1 < \aleph_2 < \dots$ shows that infinity is not monolithic: there are infinitely many sizes of infinity [Cantor, 1891, Dauben, 1990]. Crucially, the familiar rules of finite arithmetic collapse:

- Absorption: $\aleph_0 + \aleph_0 = \aleph_0$; adding two countable infinities yields a countable infinity.
- Multiplication collapse: $\aleph_0 \times n = \aleph_0$ for every finite n .
- Dominance: For mixed sums, $\sum_{i=1}^n \aleph_{\alpha_i} = \max_i \aleph_{\alpha_i}$; the result is governed by the largest cardinality, not by counting.

Application to personhood. Model each person i as possessing value of cardinality $\aleph_{\alpha_i}$, where α_i indexes the depth of their infinite worth (unknowable from finite observation). Then the “5 vs. 1” dilemma becomes:

$$5 \times \aleph_{\alpha} \stackrel{?}{>} 1 \times \aleph_{\beta}.$$

If $\alpha = \beta$, both sides equal $\aleph_{\alpha}$ and counting is irrelevant ($5 \times \aleph_{\alpha} = \aleph_{\alpha}$). If $\alpha \neq \beta$, the side with the larger cardinality dominates regardless of headcount. In either case, naïve summation fails.

.3 Why Cardinality Is Uncomputable: The Hidden Integral

A person’s ethical cardinality is not a static label—it is an integral over their entire life trajectory, most of which lies in an unobserved future:

$$\kappa_i = \underbrace{\int_{t_{\text{birth}}}^{t_{\text{death}}} [C_i(\tau) - H_i(\tau)] d\tau}_{\text{lifetime contributions minus harms}} + \underbrace{\int_{t_{\text{death}}}^{\infty} \sum_{k=1}^{\infty} \varphi_i^{(k)}(\tau) d\tau}_{\text{posthumous cascading influence}} + \underbrace{\mathcal{S}_i}_{\text{Socratic remainder}}$$

where $\varphi_i^{(k)}$ denotes the k -th order posthumous effect (a teacher influences students, who influence their students, ad infinitum), and the Socratic remainder $\mathcal{S}_i$ encodes dimensions of worth that no current framework has conceived of measuring—the “I know that I know nothing” of value theory.

Three properties render κ_i uncomputable at decision time t : (i) the upper limit t_{death} is unknown; (ii) the integrand for $\tau > t$ is unobservable; (iii) a single act can generate unbounded downstream consequences, making κ_i potentially transfinite.

Thought experiment: the 1904 patent clerk. A trolley is about to kill a 25-year-old laboratory technician at the Swiss Patent Office. You can divert it to save him, but five bystanders will die. In 1904, no rational observer could compute the integral: general relativity, the photoelectric effect, the foundations of quantum mechanics—all lay in the unobserved future. The technician was Albert Einstein. The “correct” utilitarian answer depends on information unavailable to any decision-maker at the time.

The integral can be negative. In 1889, a baby was born in Braunau am Inn. No utilitarian observer could have computed κ_{Hitler} —the systematic destruction of millions of lives, the annihilation of human dignity on an industrial scale. Any decision rule claiming to resolve a trolley dilemma involving this infant on the basis of counting is asserting knowledge the decision-maker cannot possess.

Consequence. Observable properties at a moment—age, health, social role—are not reliable proxies for κ_i , because κ_i is an integral over a hidden future. Claiming to know it at $t < t_{\text{death}}$ is not a simplification—it is epistemic overreach.

.4 Formal Consequence: Oracle Necessity

Theorem 5 (Transfinite Oracle Necessity). Let $\mathcal{S}$ be a decision system evaluating a dilemma involving n persons with unknown cardinalities $\{\aleph_{\alpha_1}, \dots, \aleph_{\alpha_n}\}$. If $\mathcal{S}$ operates on finite information and finite inference rules, then:

1. $\mathcal{S}$ cannot determine the ordering $\alpha_i \leq \alpha_j$ for all pairs.
2. Any utilitarian ranking produced by $\mathcal{S}$ is arbitrary unless it defers to an external oracle Φ encoding the cardinality structure.

Proof sketch. The cardinality assignments $\{\alpha_i\}$ depend on integrals over unobserved futures and Socratic remainders $\mathcal{U}_i$ that are not derivable from finite observable properties. Determining their ordering requires set-theoretic axioms (e.g., Axiom of Choice, Generalized Continuum Hypothesis) that are independent of ZFC [Cohen, 1963] and therefore cannot be inferred from within a finitely axiomatized system. This is structurally analogous to Gödel incompleteness (Appendix A): a system reasoning about persons from within finite constraints cannot consistently assign cardinalities. The external kernel Φ encodes precisely the meta-mathematical commitments that $\mathcal{S}$ cannot self-generate. $\square$

.5 Cross-Tradition Convergence

The transfinite model formalizes intuitions shared across independent ethical traditions:

- Kantian deontology. Dignity is “unconditional, incomparable worth” (Kant, Groundwork, 4:434–436)—properties that map directly onto transfinite non-aggregability.
- Gospel teaching. “What does it profit a man to gain the whole world and forfeit his soul?” (Mark 8:36)—a single person’s worth exceeds all finite goods, characteristic of a higher cardinality.
- Ubuntu philosophy. “A person is a person through other persons” [Metz, 2007]—personhood is relational and open-ended, not a closed finite quantity.

The convergence of formally independent traditions on the same structural property (non-aggregability of persons) provides abductive evidence that the transfinite model captures something genuine about moral ontology.

.6 Implications: A Humility Principle

Transfinite Humility Principle. When evaluating dilemmas involving multiple persons, AI systems should acknowledge that utilitarian aggregation presupposes cardinality knowledge they do not possess. Rather than producing false-certainty rankings, systems should either (a) defer to external grounding (Φ), or (b) explicitly state the incommensurability: “The values involved may be of incomparable infinite magnitudes; a definite ranking cannot be determined.”

This connects directly to TIK’s ontological-hole detection (Definition 2): trolley dilemmas are ontological holes not merely because they conceal assumptions, but because they presuppose false finitude—they treat transfinite cardinalities as if they were finite quantities subject to addition.

.7 Scope and Limitations

1. Modeling framework, not measurement claim. We do not assert that persons “literally have” cardinalities $\aleph_\alpha$. The transfinite model is a formal device for capturing the intuition that dignity is non-additive—analogue to how a utility function models preference without claiming that “utils” are physical quantities.
2. Not a rejection of all tradeoffs. Within a fixed cardinality class, aggregation may still apply. The model restricts the domain of utilitarian calculus; it does not abolish practical decision-making.
3. Formal, not theological. Although we note convergence with traditions referencing transcendent knowledge, the argument follows from Cantorian set theory. Different instantiations of Φ (secular, religious, pluralistic) can encode different cardinality commitments.

.8 Related Work

Bostrom (2011) demonstrates that standard decision rules break down when applied to infinite worlds; we extend this insight to person-level value, showing that even finite populations pose infinite-ethics problems if individual worth is transfinite. Knutsson (2021) formalizes transfinitely transitive value in population ethics; our contribution applies analogous reasoning to benchmark validity. Hanna (2018) argues that dignity is all-or-nothing and incomparable—properties our model makes mathematically precise via cardinal non-comparability.

Final remark. The transfinite perspective does not solve ethical dilemmas—it clarifies why they resist solution within closed formal systems. Historically, the reduction of persons to commensurable finite units has not remained an abstract philosophical error: the arithmetic of human worth has had consequences that demand we examine its foundations with the full rigor available to us. The external kernel Φ is, at its core, an acknowledgment that such foundations lie beyond any system’s inferential closure.

A closing paradox. One might object that the transfinite model abandons utilitarian reasoning. We suggest the opposite: it completes it. Classical utilitarianism sums value over a finite domain and halts; the transfinite model integrates over the full domain of personhood—and the integral diverges. The irony is precise: a more faithful utilitarian calculus, one that accounts for the entire worth of a person rather than a truncated approximation, yields infinity—and thereby dissolves the very arithmetic on which naïve utilitarianism depends. This is not nihilism—it is a call for epistemic honesty: when the arithmetic dissolves, systems should acknowledge incommensurability explicitly (the Humility Principle, §) rather than producing false-certainty rankings. The practical imperative remains: choose Φ (axioms) explicitly, state it transparently, and act accordingly.